

HISTOLOGY OF INTEGUMENTARY

System

Introduction

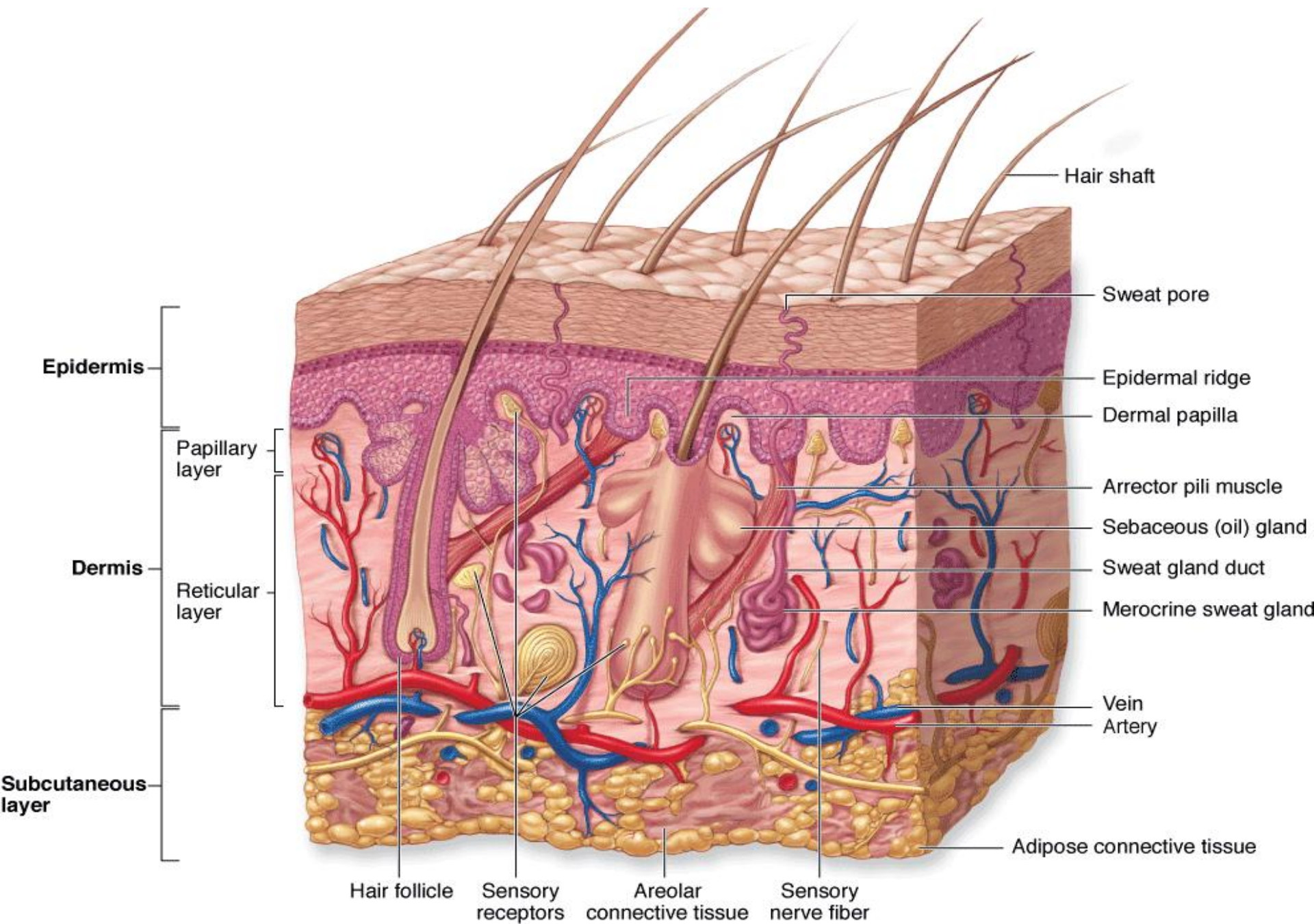


- Integumentary system is composed of :
 - Skin
 - **Epidermal derivatives :**
 - Hairs
 - Nails
 - Sebaceous and Sweat Glands

Skin: introduction



- The skin:
 - ▣ is the largest single organ of the body
 - ▣ typically accounting for 15–20% of total body weight and, in adults
 - ▣ presents 1.5–2 m² of surface to the external environment.
 - ▣ Also known as the integument (L. integumentum, covering) or cutaneous layer.



The functions of the skin



- Protection
- Sensory
- Thermoregulation
- **Metabolic**
- Sexual signaling

Epidermis

- The epidermis forms the major distinction between:
 - ▣ **thick skin**-----found on the palms and soles, and
 - ▣ **thin skin**----- found elsewhere on the body
- The designations "thick" and "thin" refer to the thickness of the epidermal layer, which varies from :
 - ▣ 75 to 150 μm -----for thin skin and
 - ▣ 400 to 1400 μm (1.4 mm)----- for thick skin

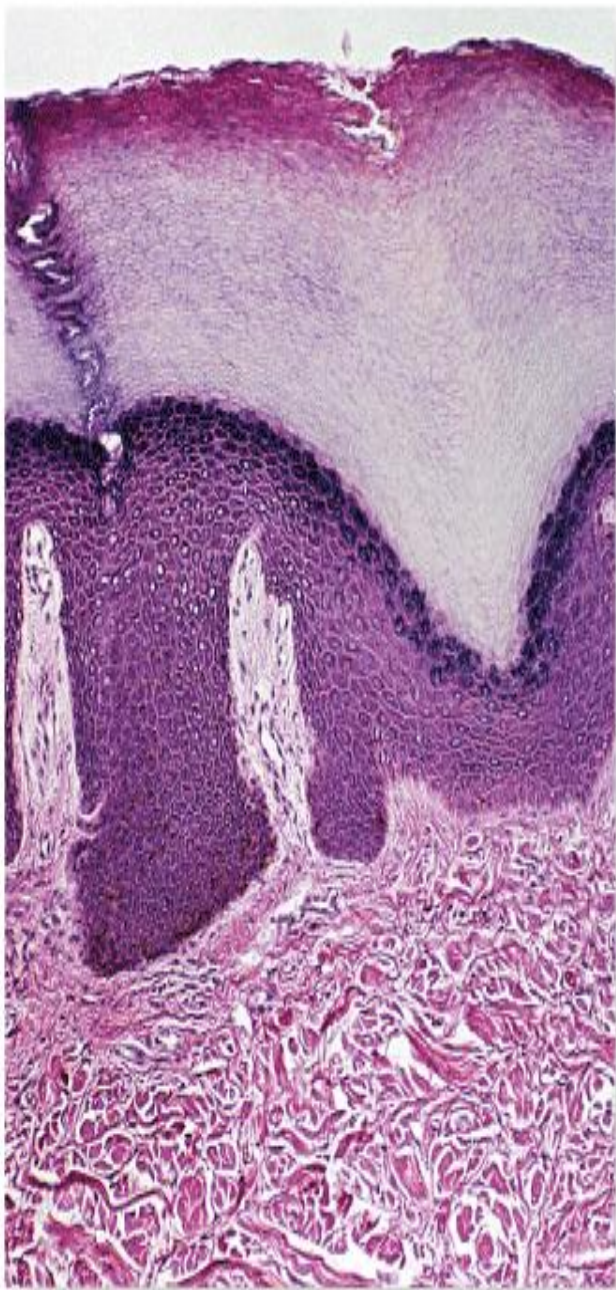
Epidermis



- Total skin thickness (epidermis plus dermis) also varies according to site.
- For example:
 - ▣ skin on the back is about----- 4 mm thick
 - ▣ Skin of the scalp is about----- 1.5 mm thick

Epidermal layers

- The epidermis consists of four layers/ five layers in thick skin/ keratinocytes :
 - The **basal layer (stratum basale)**
 - The **spinous layer (stratum spinosum)**
 - The **granular layer (stratum granulosum)**
 - The **stratum lucidum**
 - The **stratum corneum**



Stratum corneum

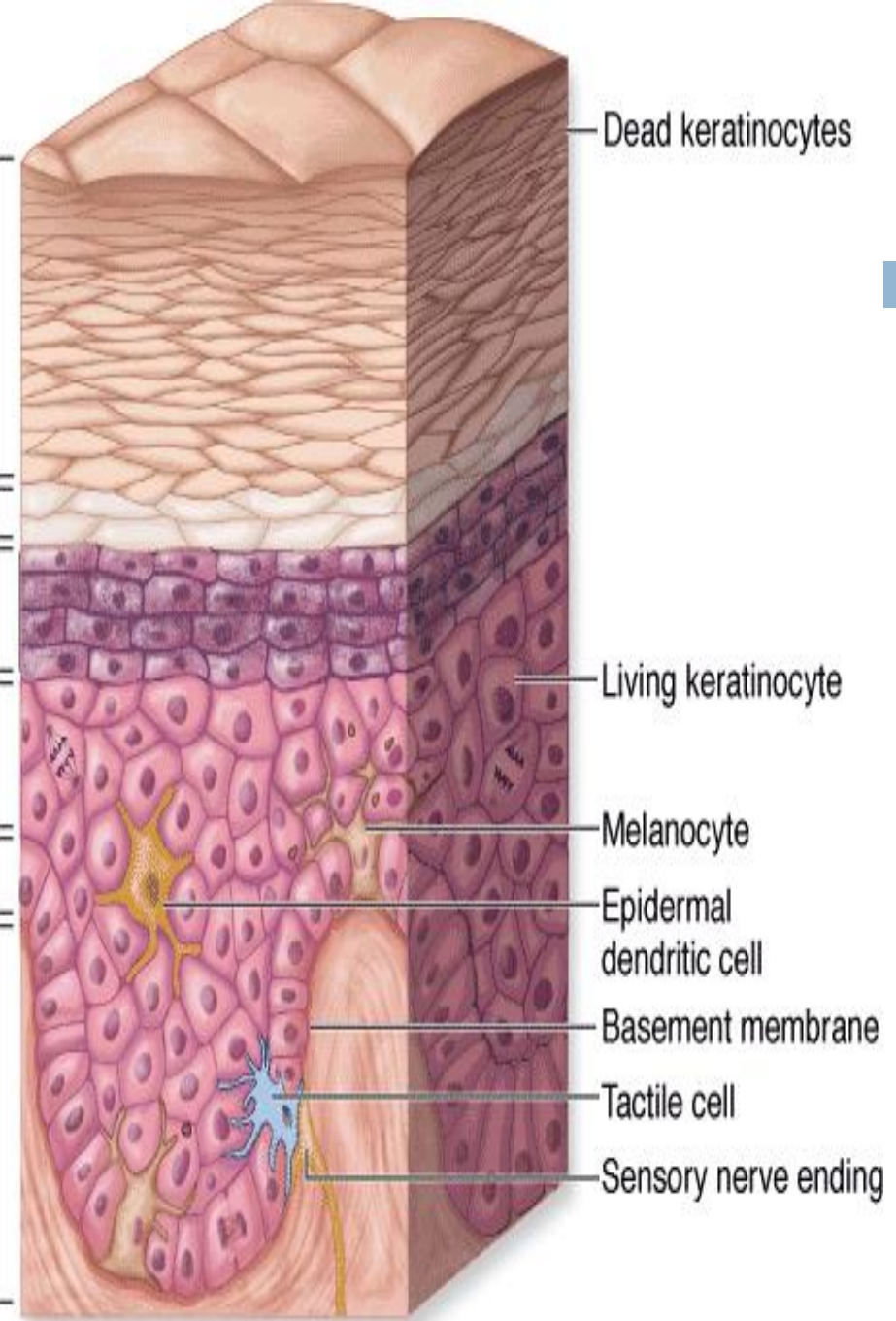
Stratum lucidum

Stratum granulosum

Stratum spinosum

Stratum basale

Dermis



Dead keratinocytes

Living keratinocyte

Melanocyte

Epidermal dendritic cell

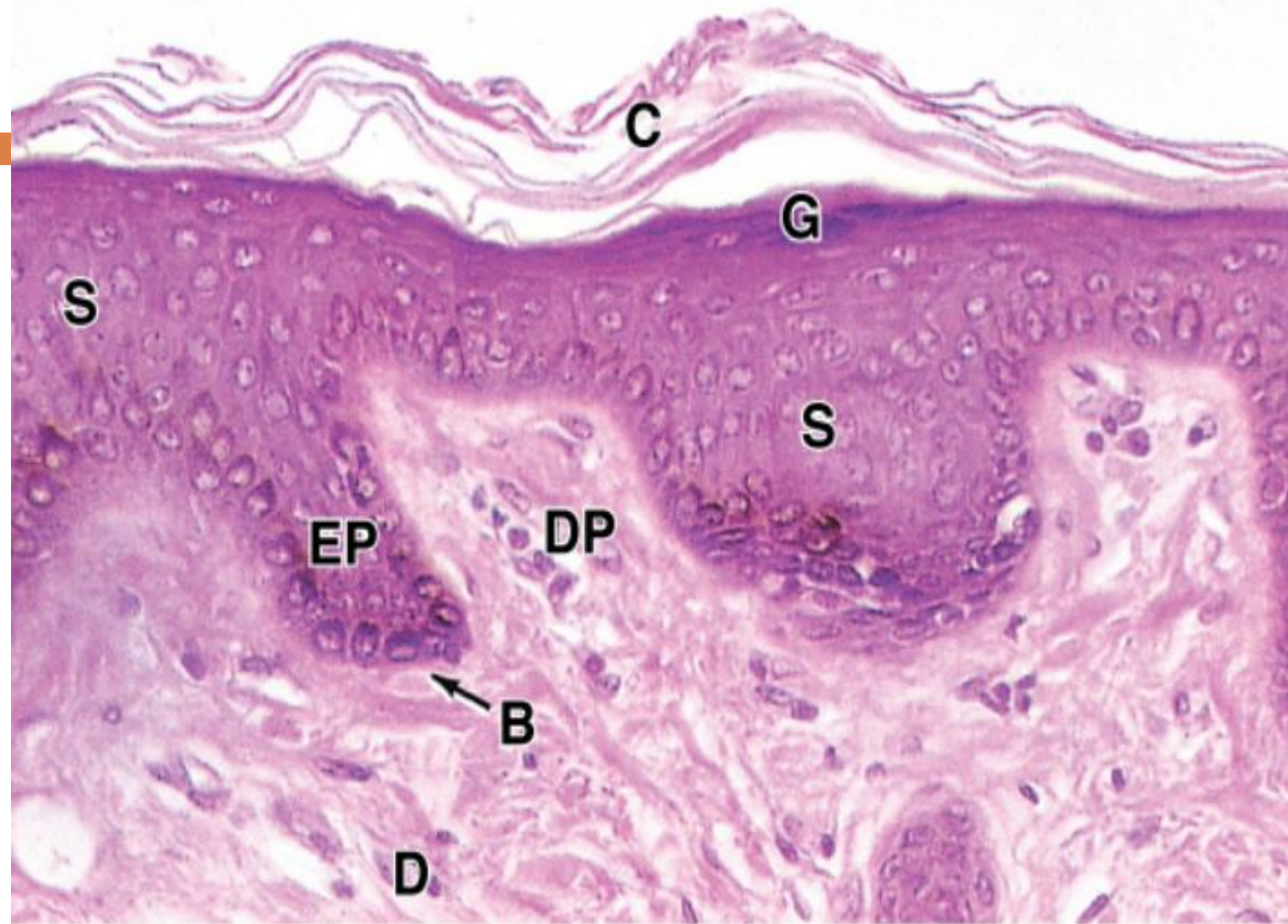
Basement membrane

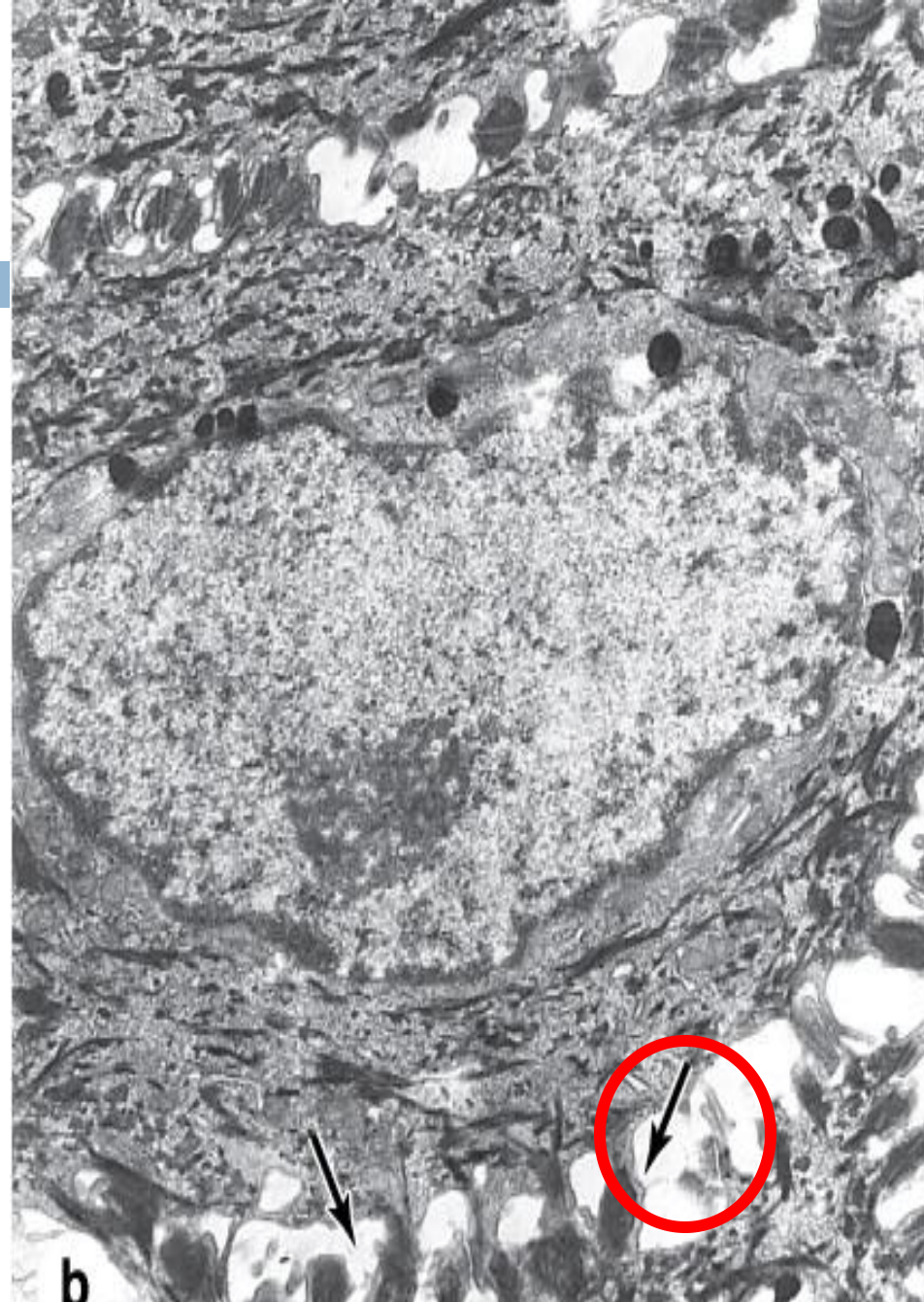
Tactile cell

Sensory nerve ending

a

b

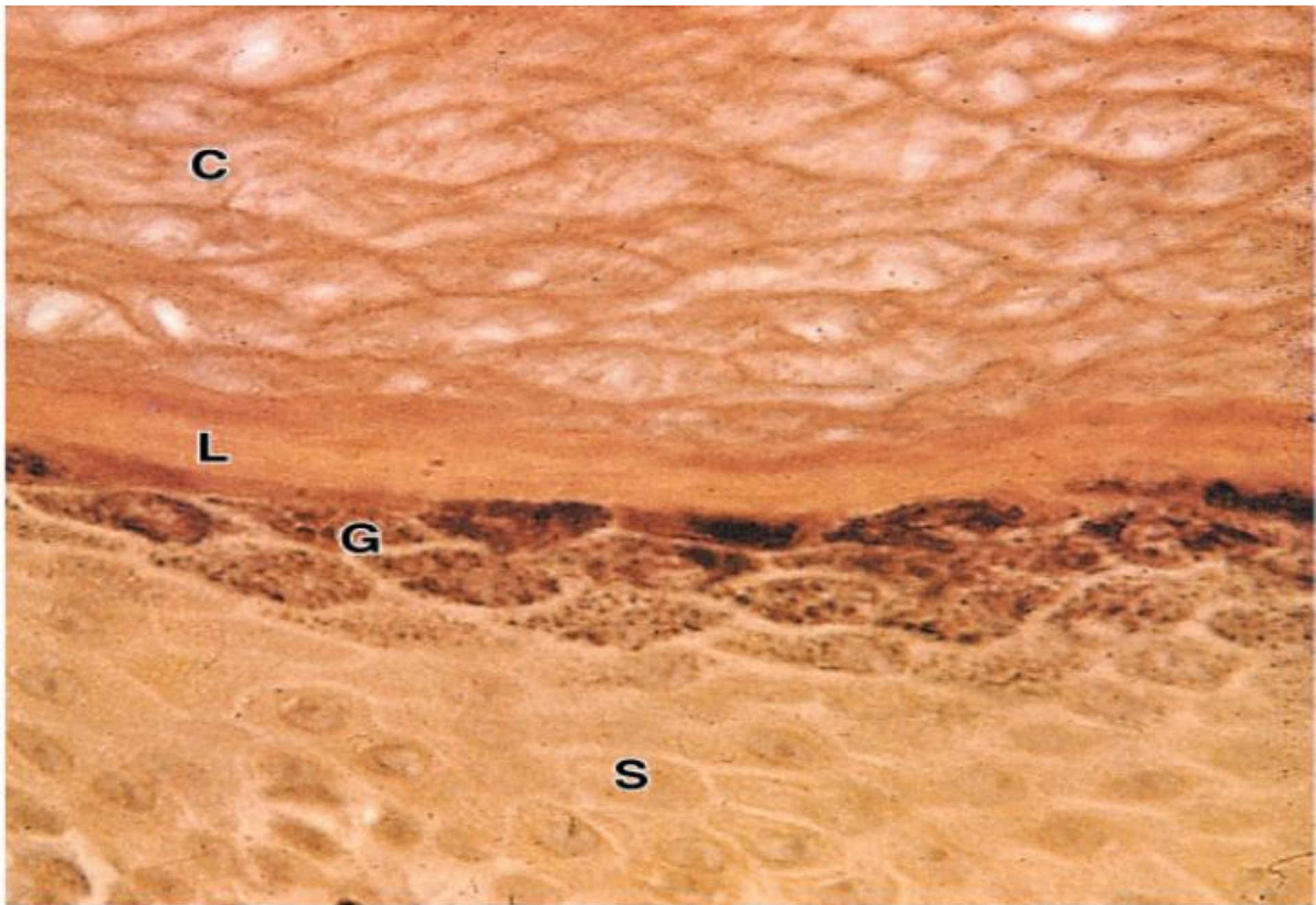




(b): TEMs show a single spinous keratinocyte with arrows marking some desmosomes at the projections. X8400.



(c): Detail of the desmosomes joining two cells showing intermediate filaments associated with desmosomes. Keratinocytes of the stratum spinosum undergo considerable protein synthesis, primarily making keratins which form large bundles called tonofibrils. These terminate at the desmosomes linking the cells (arrows) and form the short cellular extensions characteristic of this layer when the cells shrink slightly. The black granules next to the nucleus are melanin granules. X40,000.



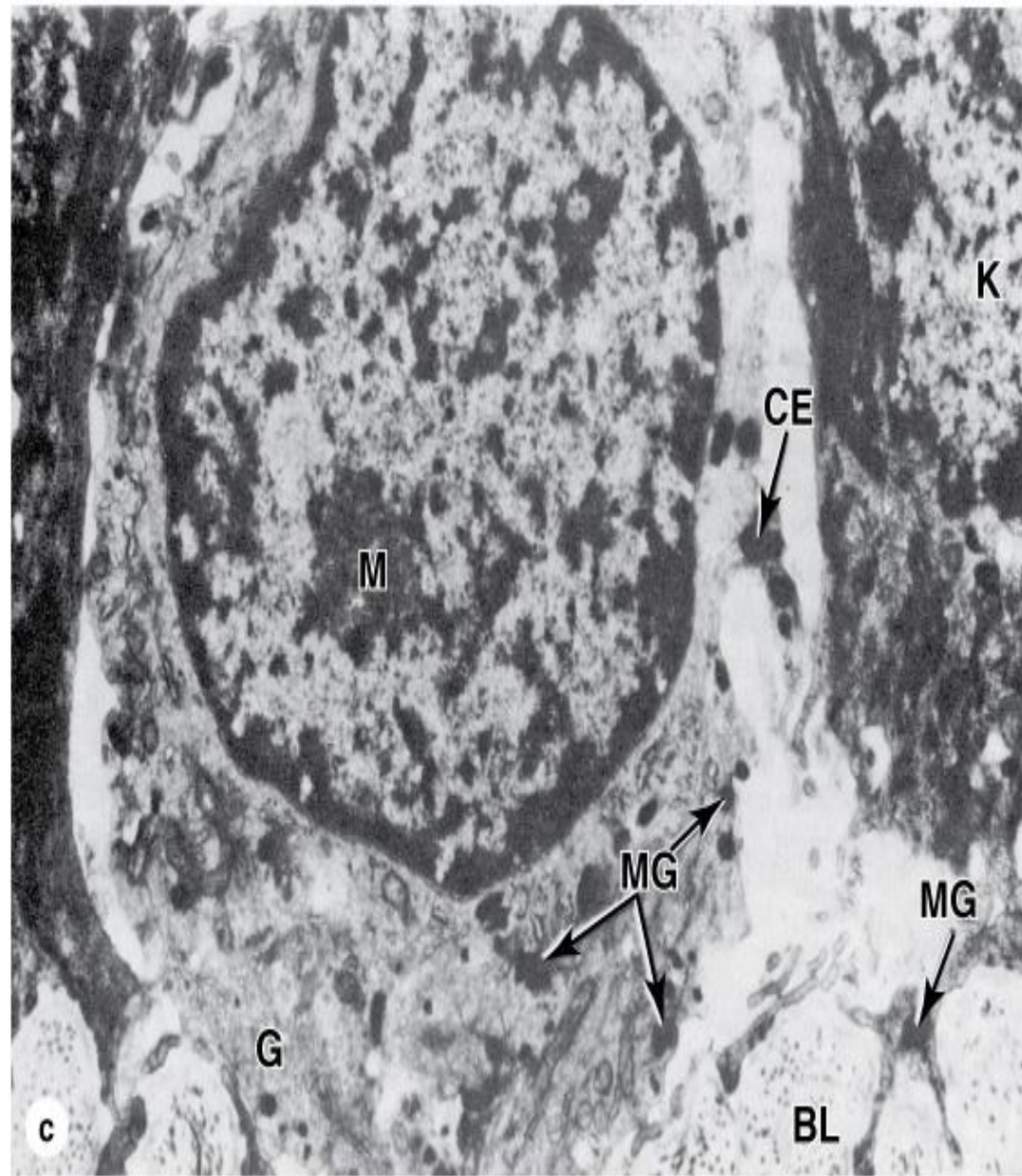
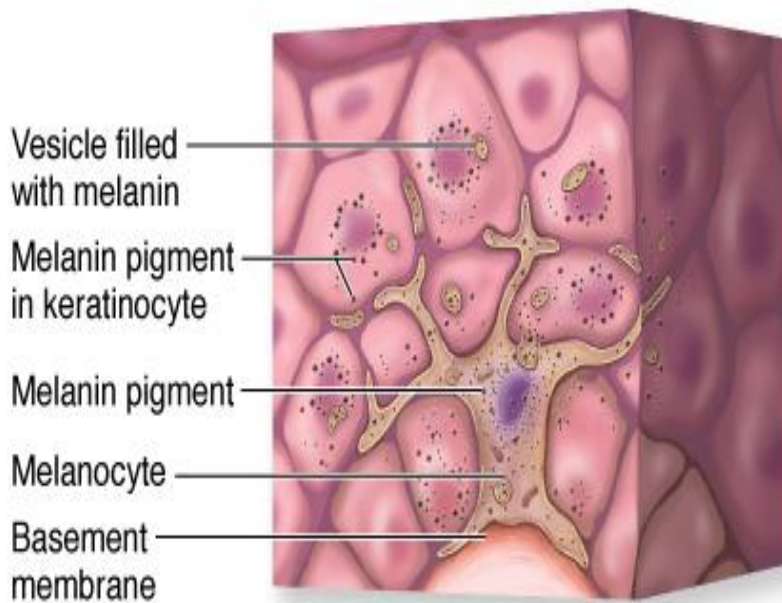
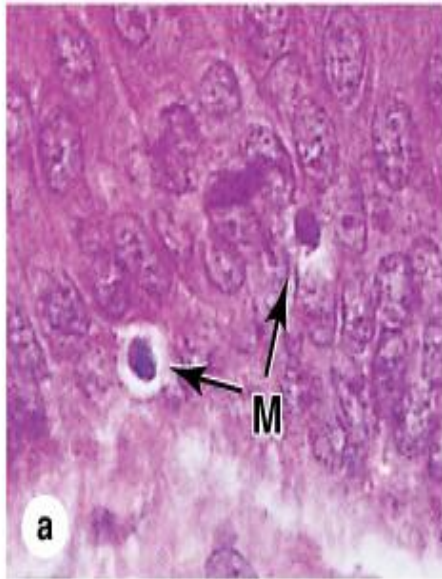
C

L

G

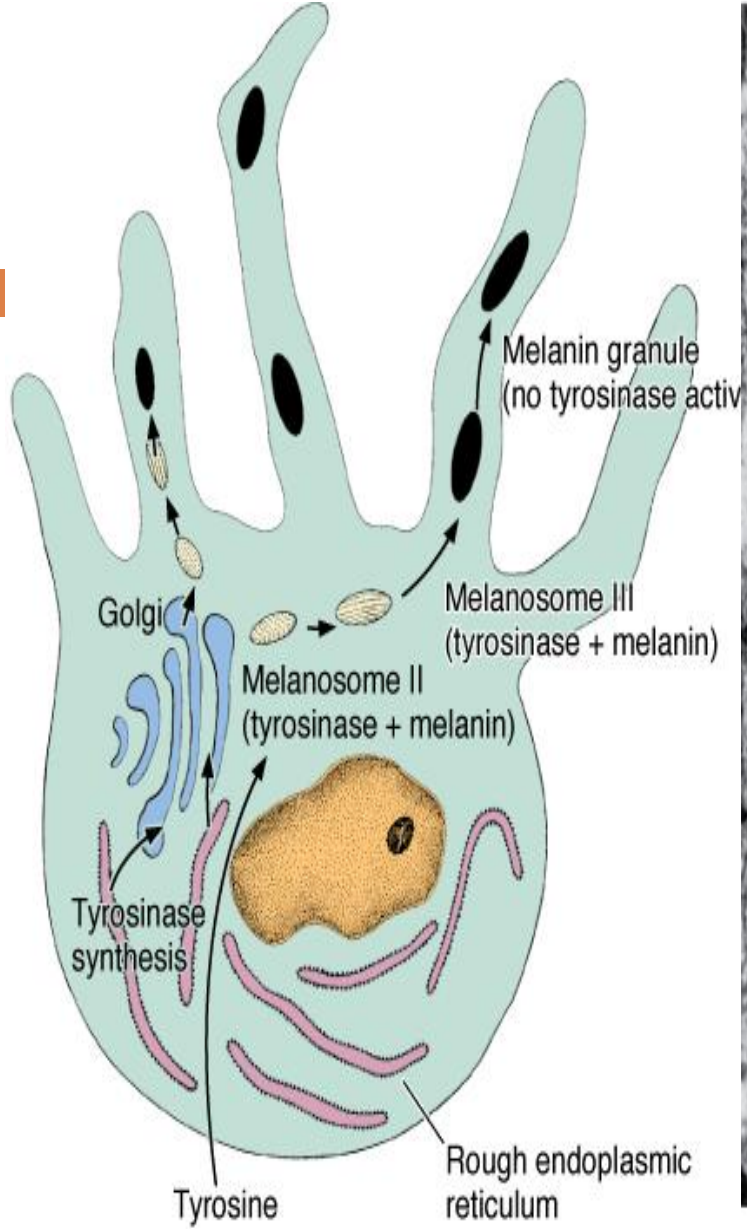
S

Stratum granulosum and stratum lucidum: thick skin.

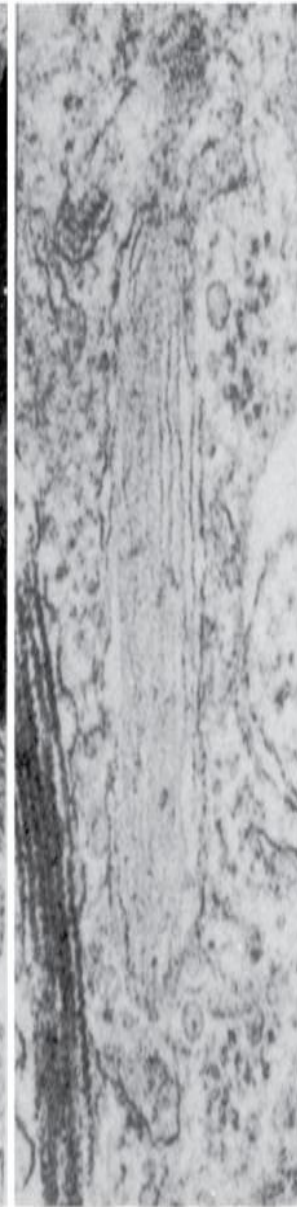


Golgi complexes (G) , Melanin granules (MG)
Cytoplasmic extensions (CE) ,Keratinocytes (K)

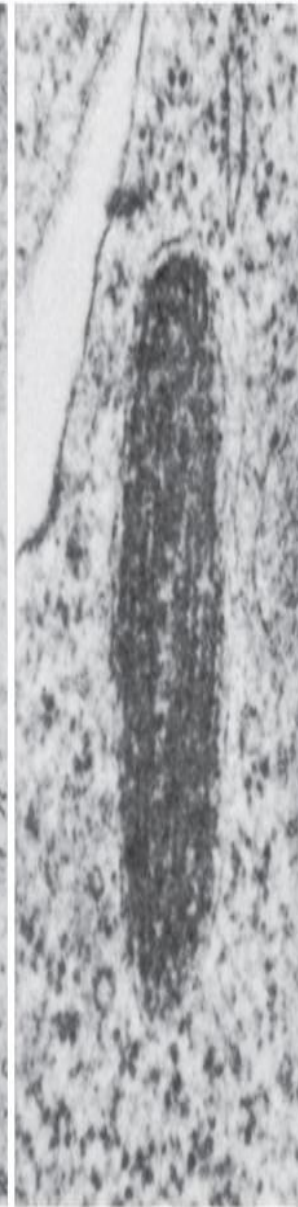
Melanocytes.



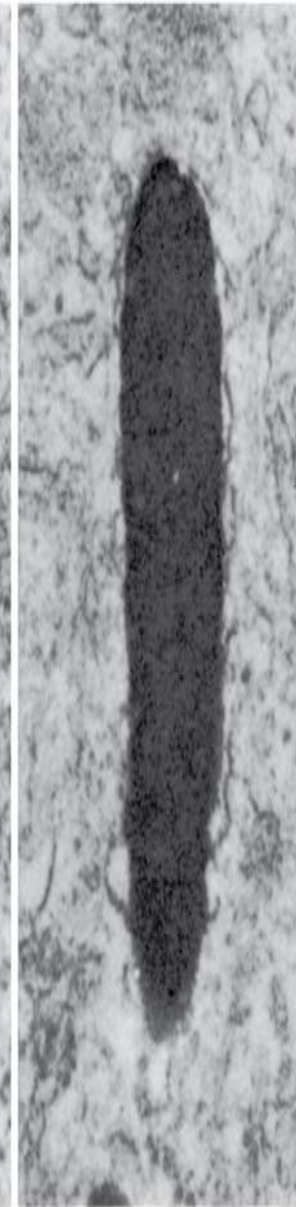
I



II



III



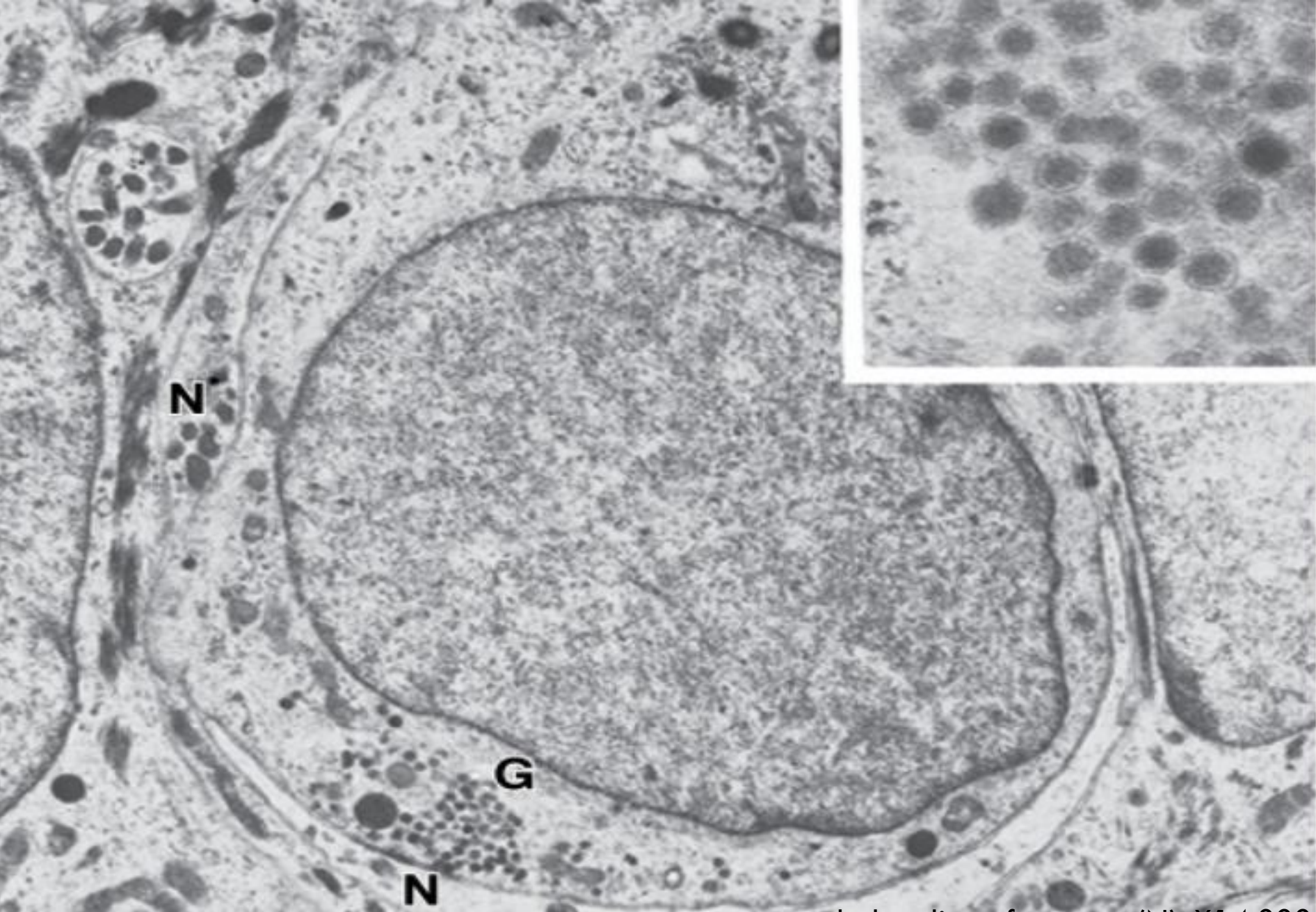
IV

Melanosome formation.





Non-segmental vitiligo of the hand.



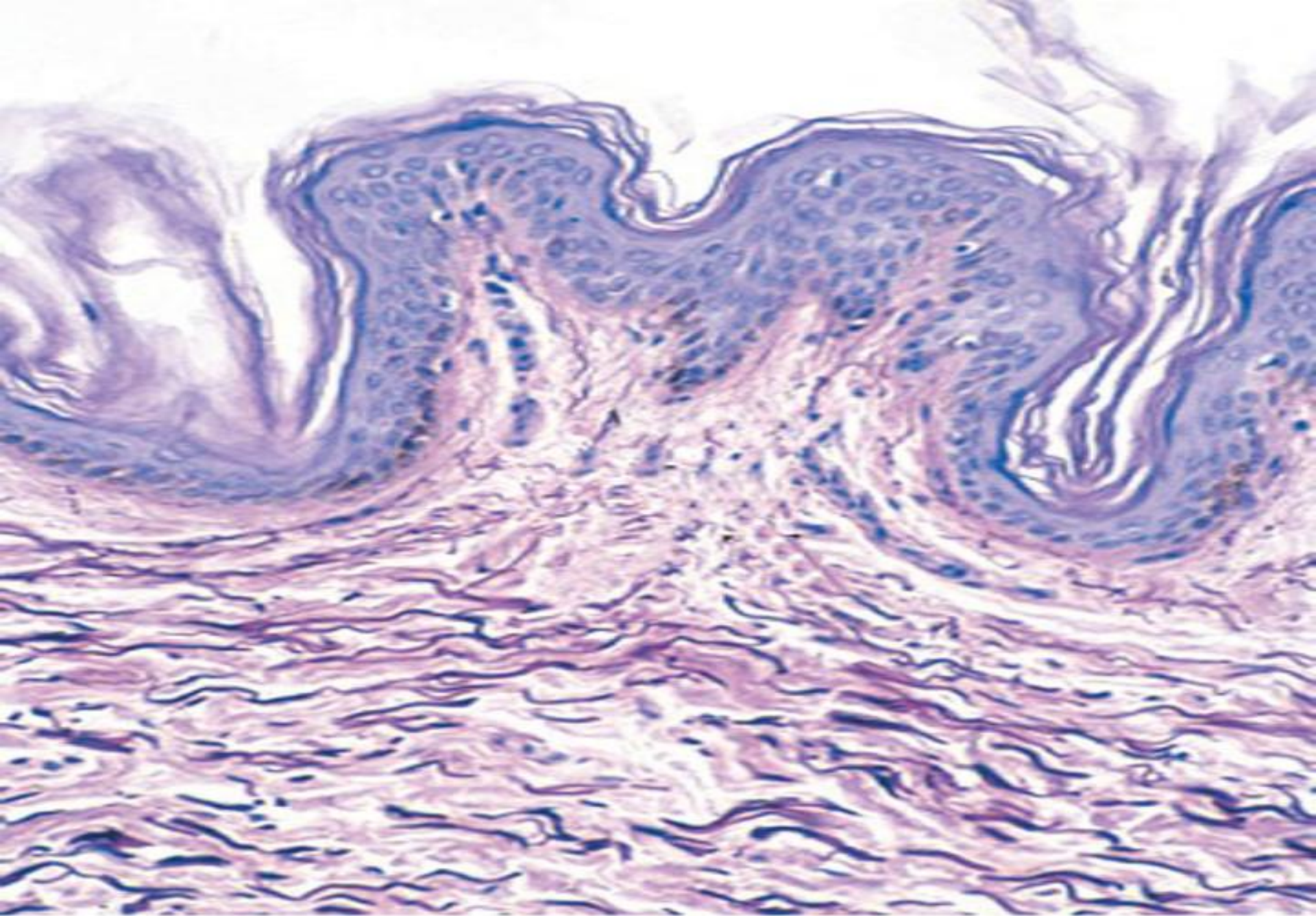
expanded ending of a nerve (N). X14,000

Tactile (Merkel) cell. dense-core cytoplasmic granules (G)

Layers of dermis



- The dermis contains two layers with rather indistinct boundaries:
 - ▣ the outermost papillary layer
 - ▣ the deeper reticular layer



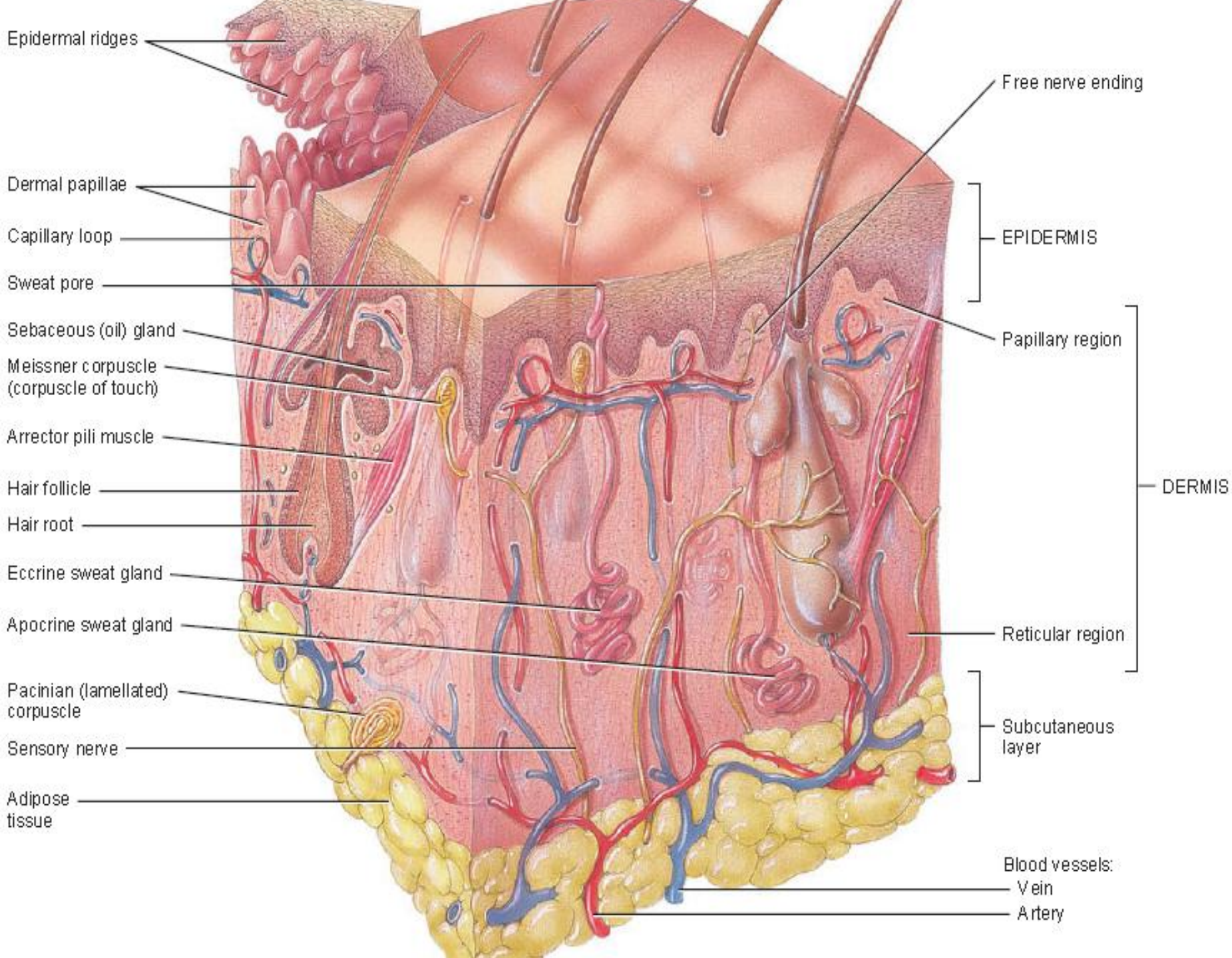
Elastic fibers of dermis.

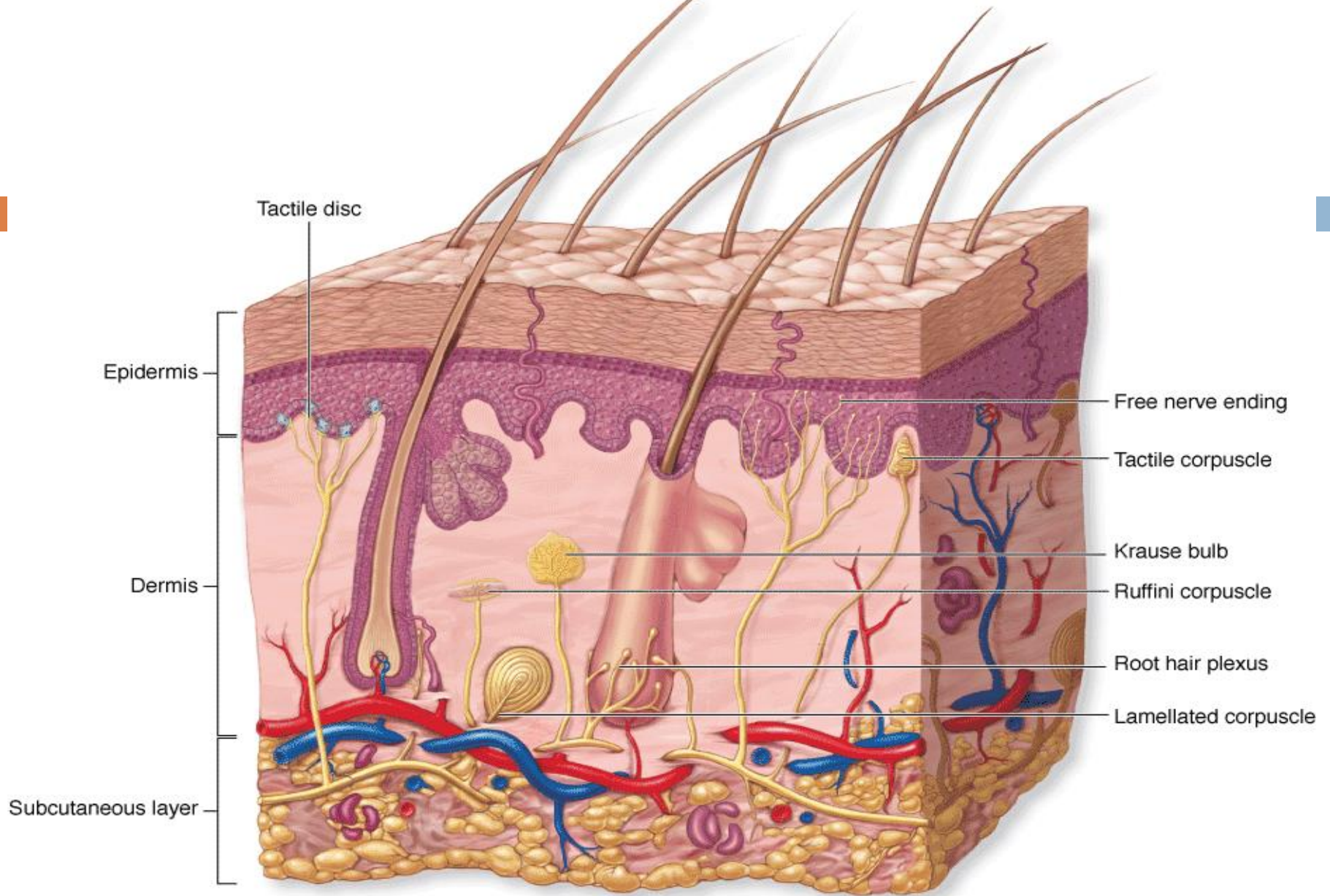
Contents of dermis

- The dermis is the site of such epidermal derivatives as the hair follicles and glands.
- There is also a rich supply of nerves in the dermis.
- The effector nerves to dermal structures:
 - ▣ are postganglionic fibers of sympathetic ganglia
 - ▣ no parasympathetic innervation is present.
- Sensory afferent nerve fibers form a network in the papillary dermis and around hair follicles, ending:
 - ▣ at epithelial tactile cells,
 - ▣ at the encapsulated sensory receptors in dermis, and
 - ▣ as free (uncovered) nerve endings among cells of the epidermis

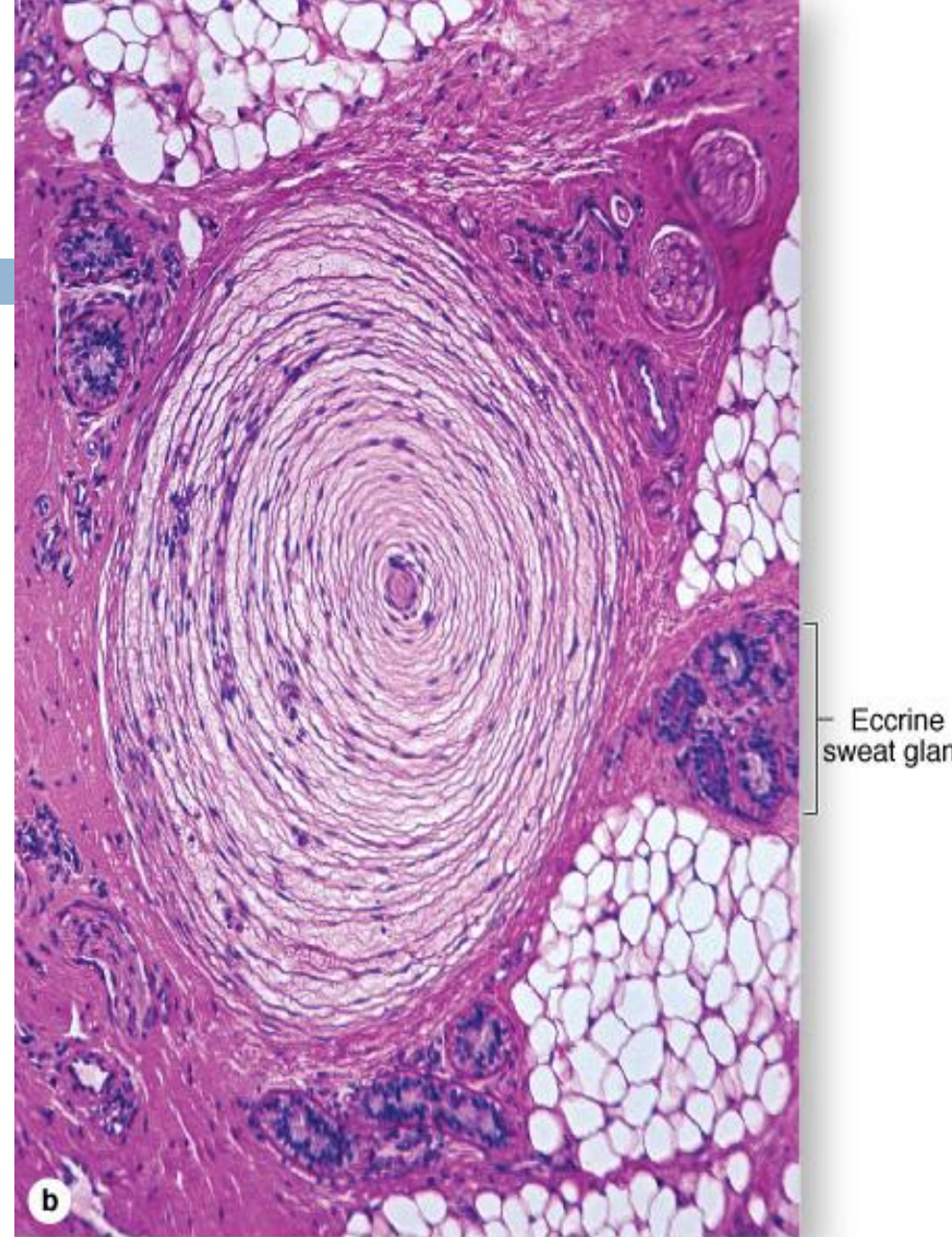
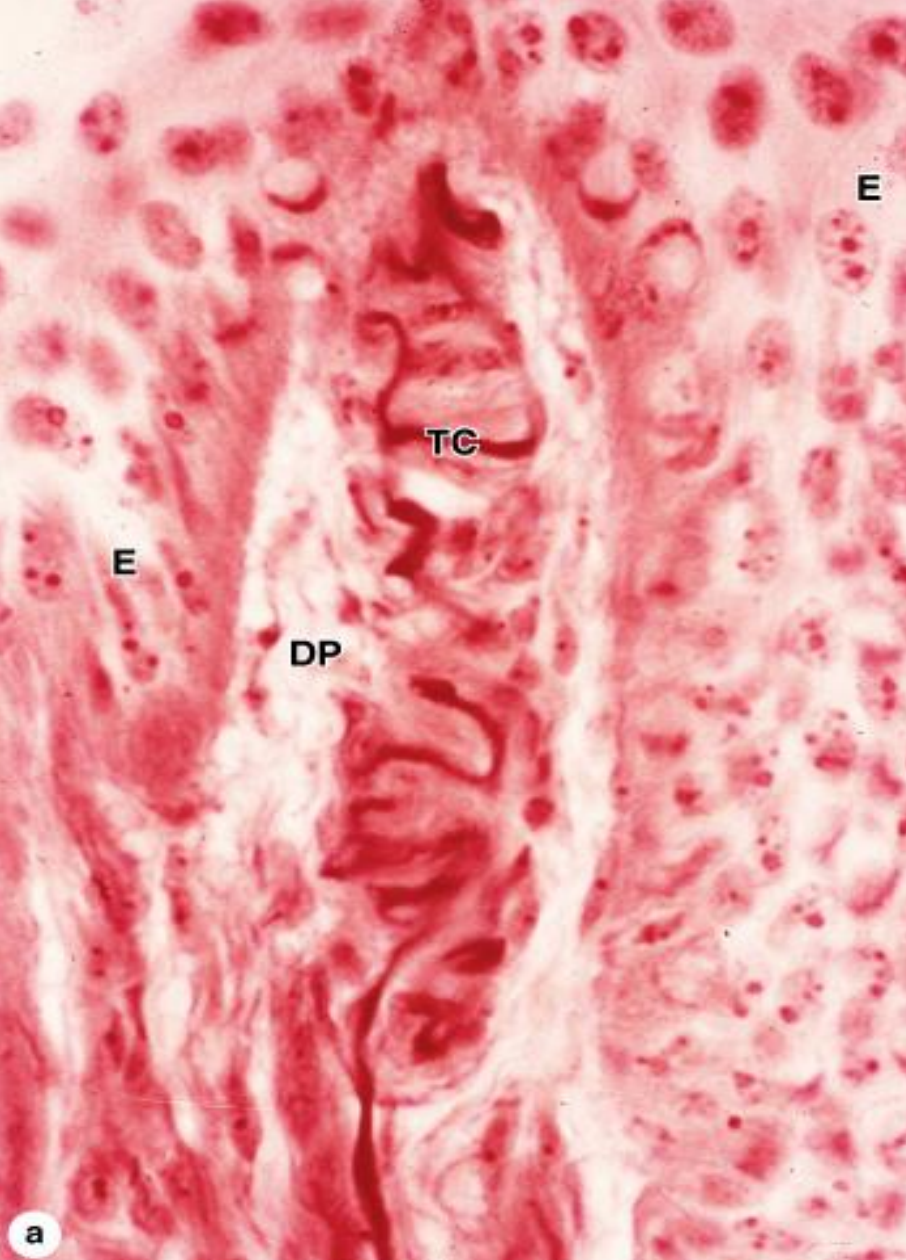
MEDICAL APPLICATION

- With age, collagen fibers thicken and collagen synthesis decreases.
- Elastic fibers steadily increase in number and thickness, so the elastin content of human skin increases approximately **fivefold** from fetal to adult life.
- In old age:
 - extensive cross-linking of collagen fibers
 - the loss of elastic fibers, and
 - degeneration of these fibers due to excessive exposure to the sun (solar elastosis)
 - cause the skin to:
 - become more fragile,
 - lose its suppleness/flexibility, and
 - develop wrinkles
- In several disorders, such as **cutis laxa and Ehlers-Danlos syndromes**:
 - there is a considerable increase in skin and ligament extensibility caused by defective **collagen-fibril processing**.





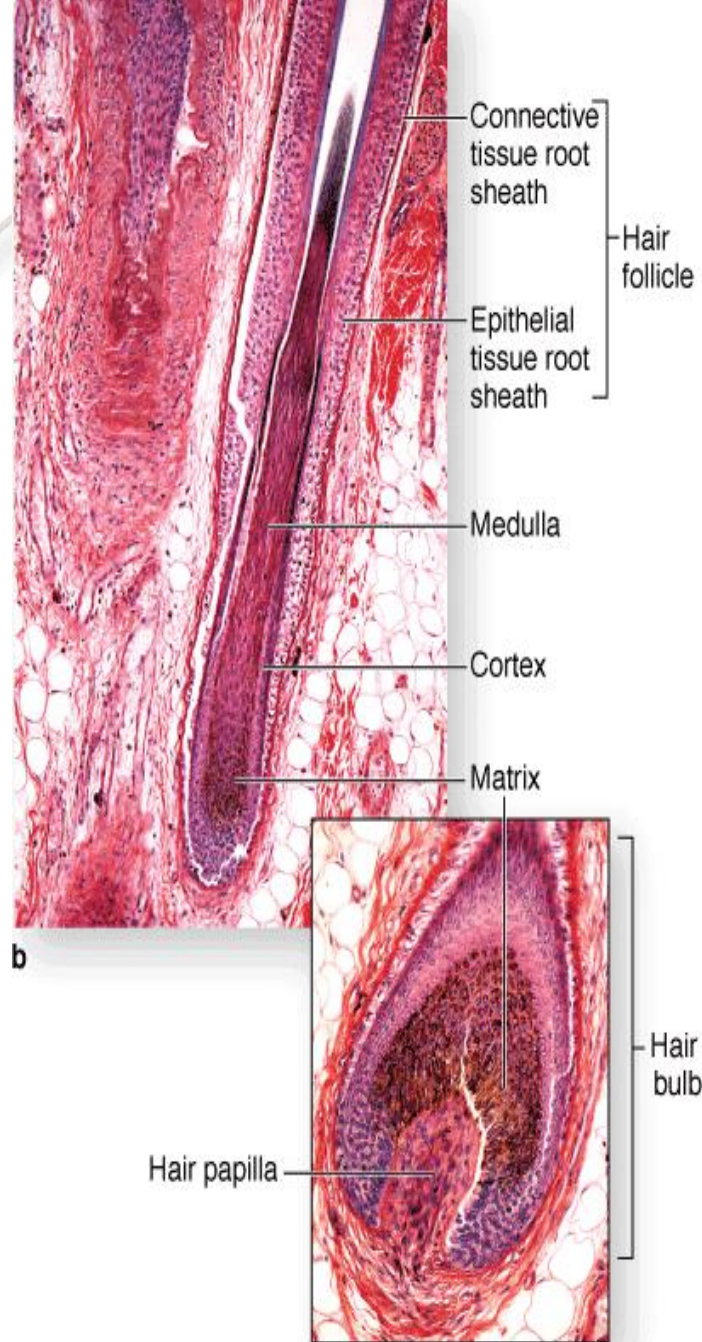
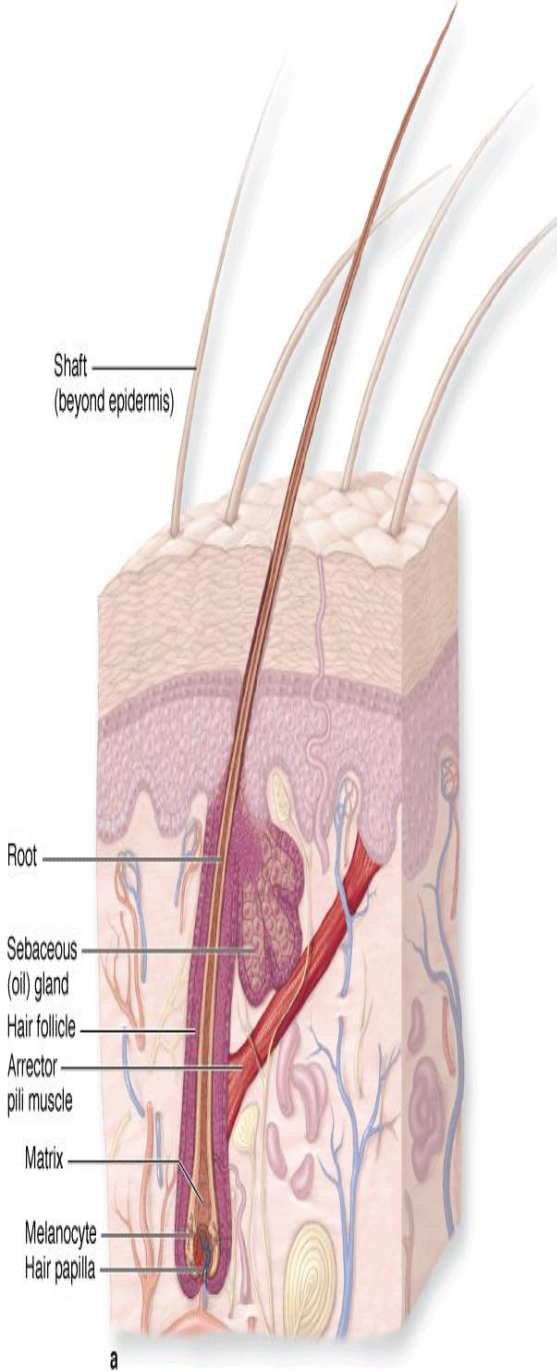
Tactile receptors.



Tactile (Meissner) corpuscle

Tactile and lamellated corpuscles.

Lamellated (Pacinian) corpuscle

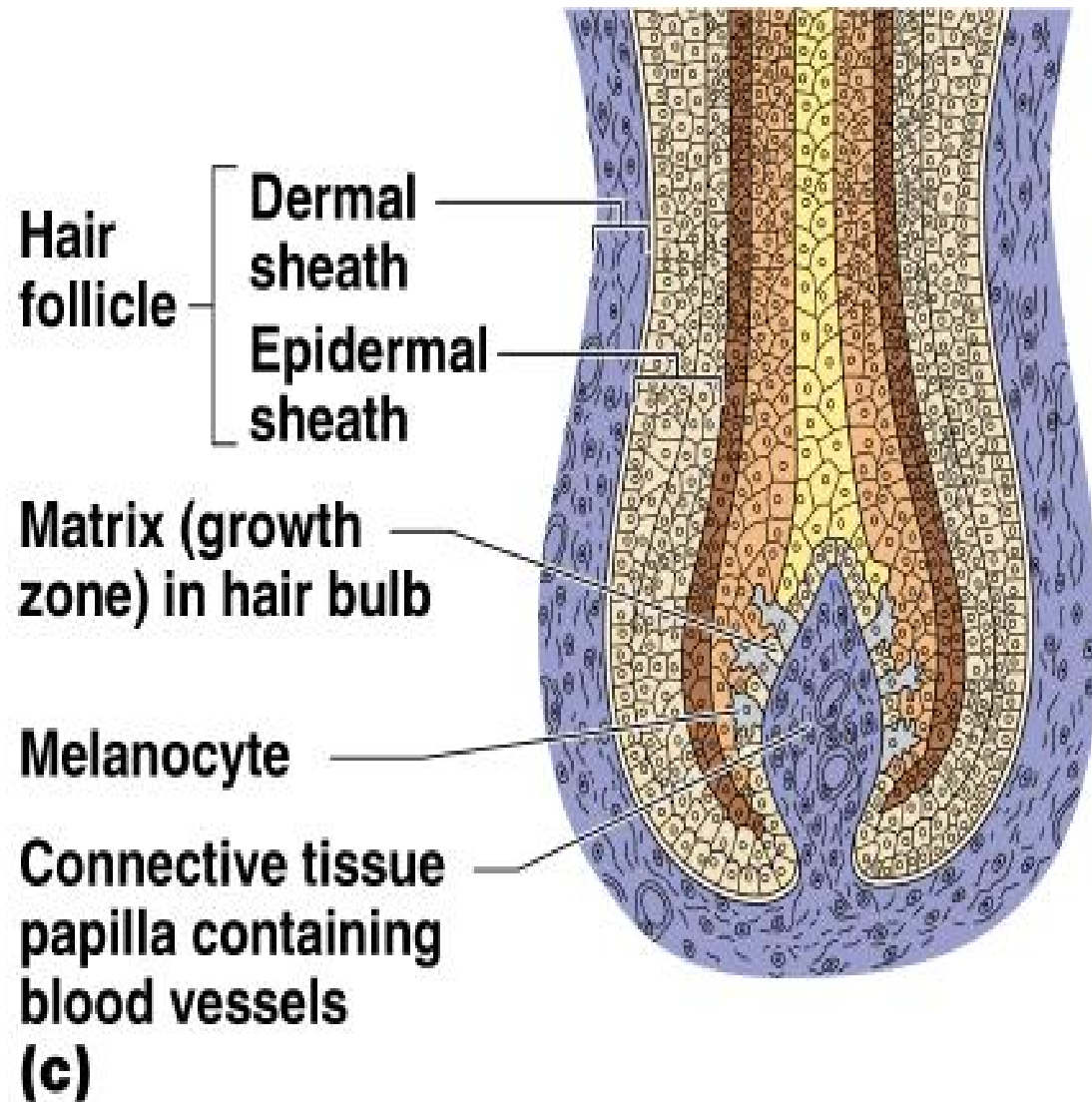


- Produced by hair bulb

- Consists of hard keratinized epithelial cells

- Melanocytes provide pigment for hair color

Hair



hair is straight and tends to be coarse



kinky or curly

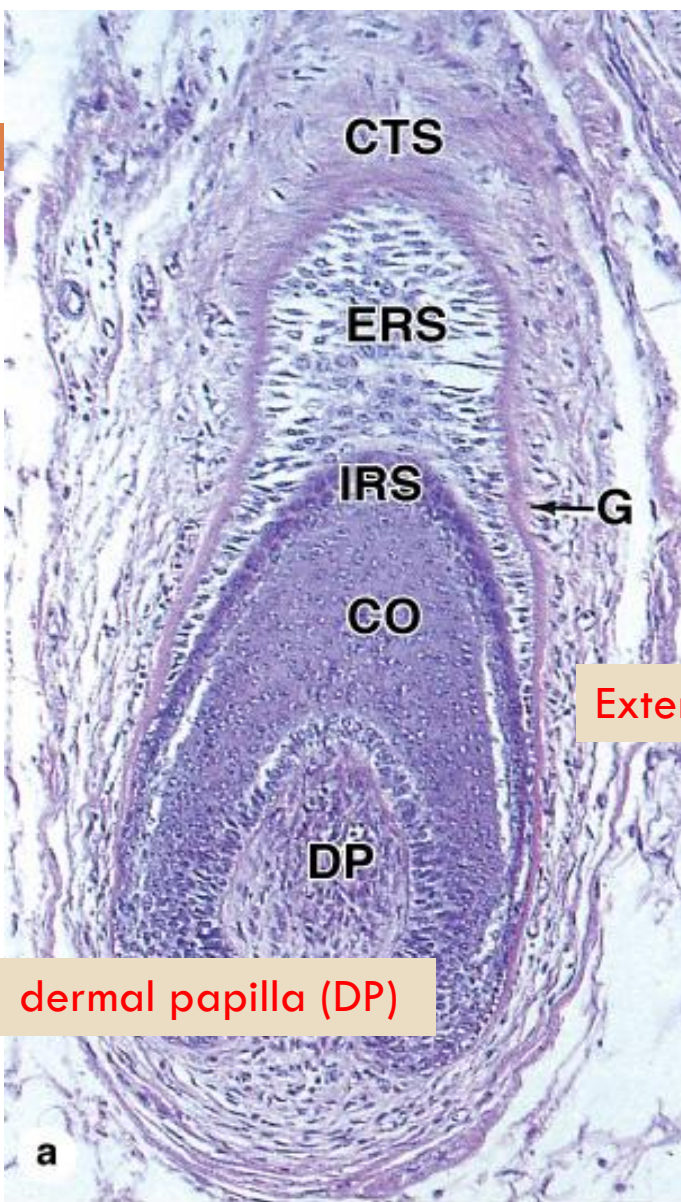


Hair is wavy

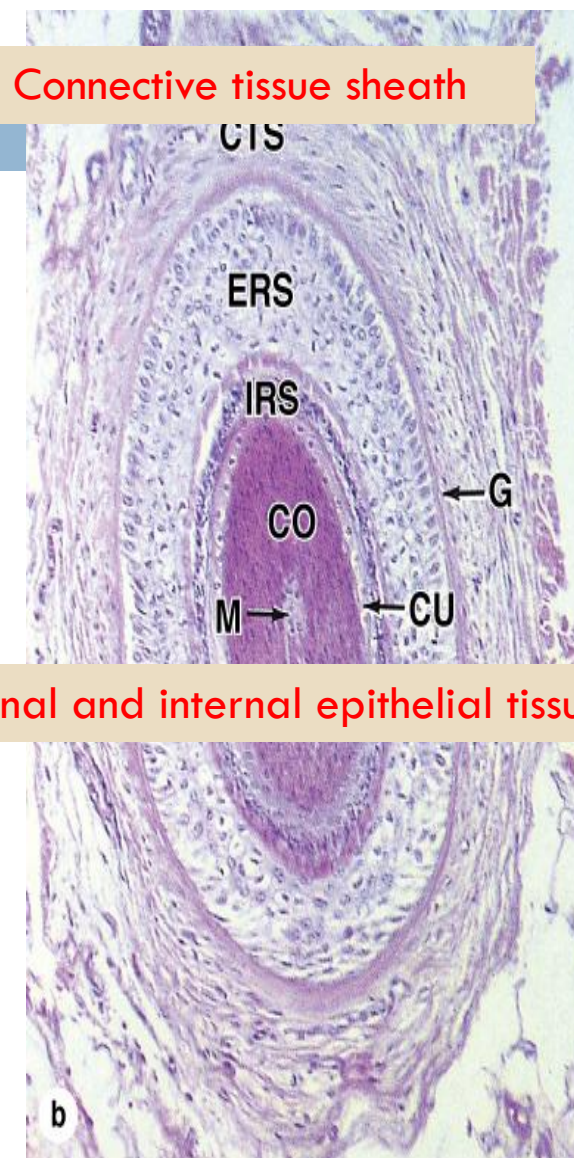


- Hirsutism = excessive hairiness
 - In women may be the result of an adrenal gland or ovarian tumor

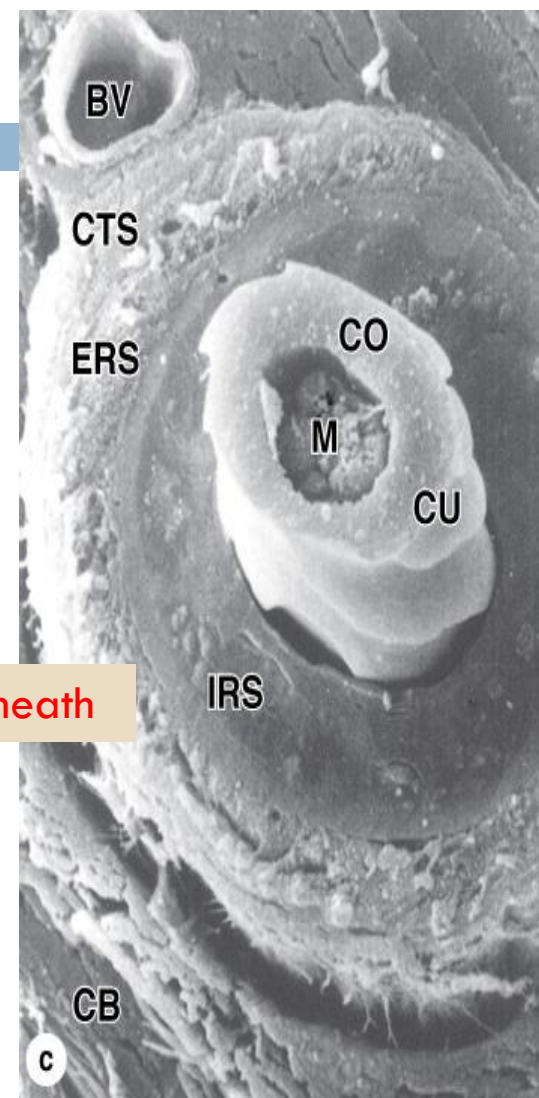
LM of a hair and its follicle near the epidermis is in transverse section. The cortex of the hair (**Co**) and internal root sheath (**IRS**), external root sheath (**ERS**), and fibrous root sheath (**FRS**) of the hair follicle are shown. The medulla of the hair shaft is not present at this level. The intensely eosinophilic cuticle (**Cu**) is made of overlapping keratinized scales of the cuticle that interlock with cells of the inner root sheath. The fibrous root sheath consists of regularly arranged dermal connective tissue. 225 $\times$. H&E.



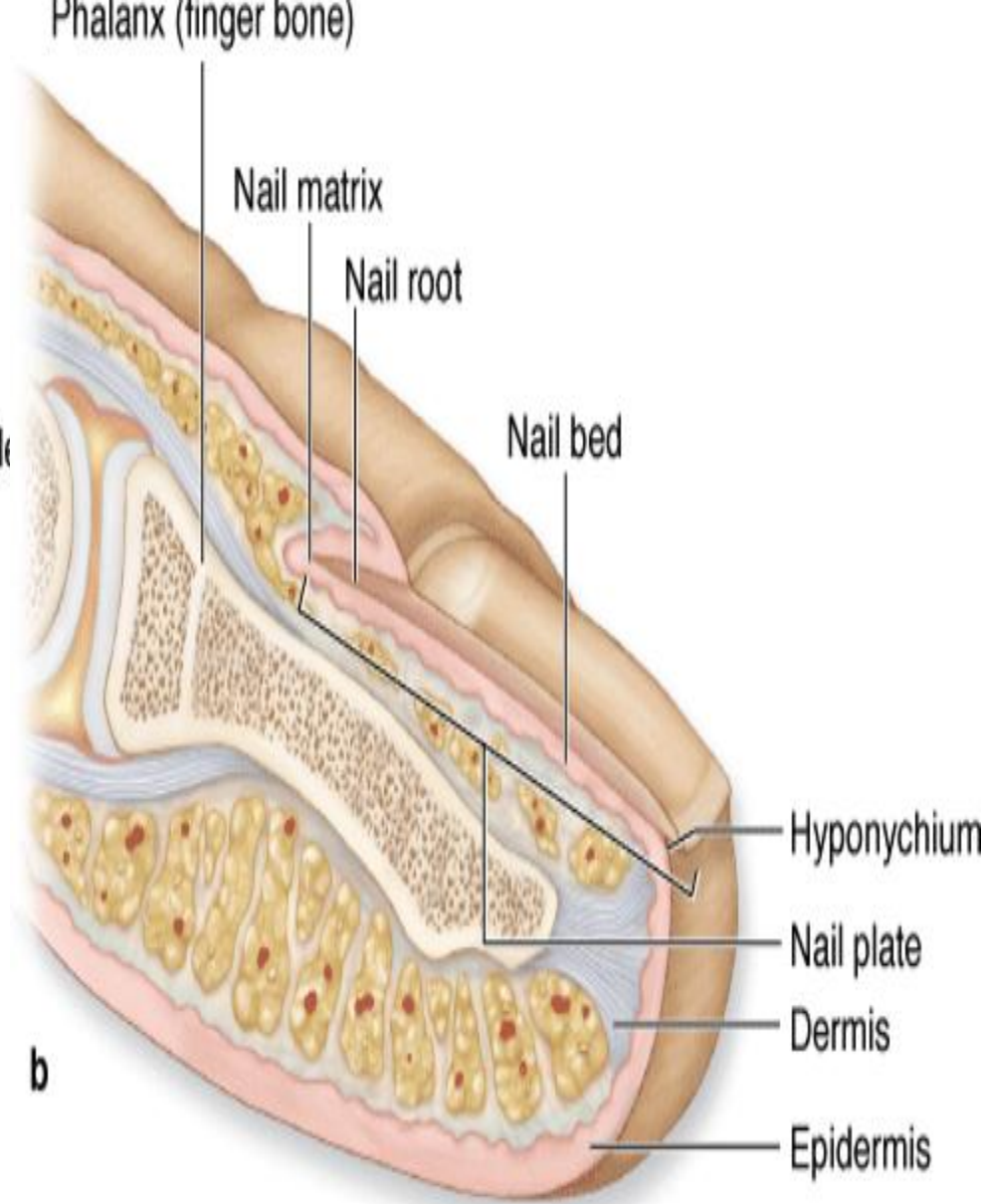
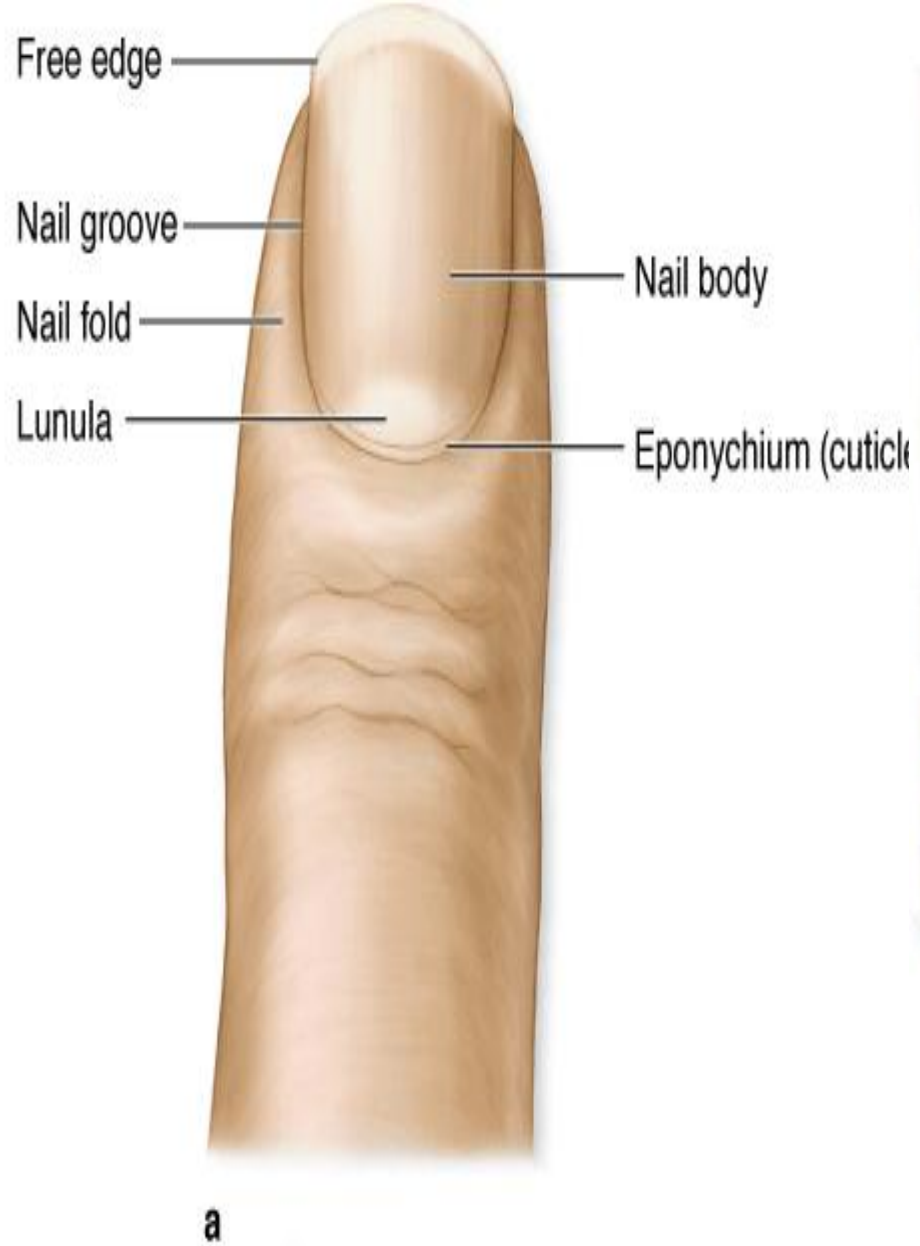
Layers of a hair and its follicle.



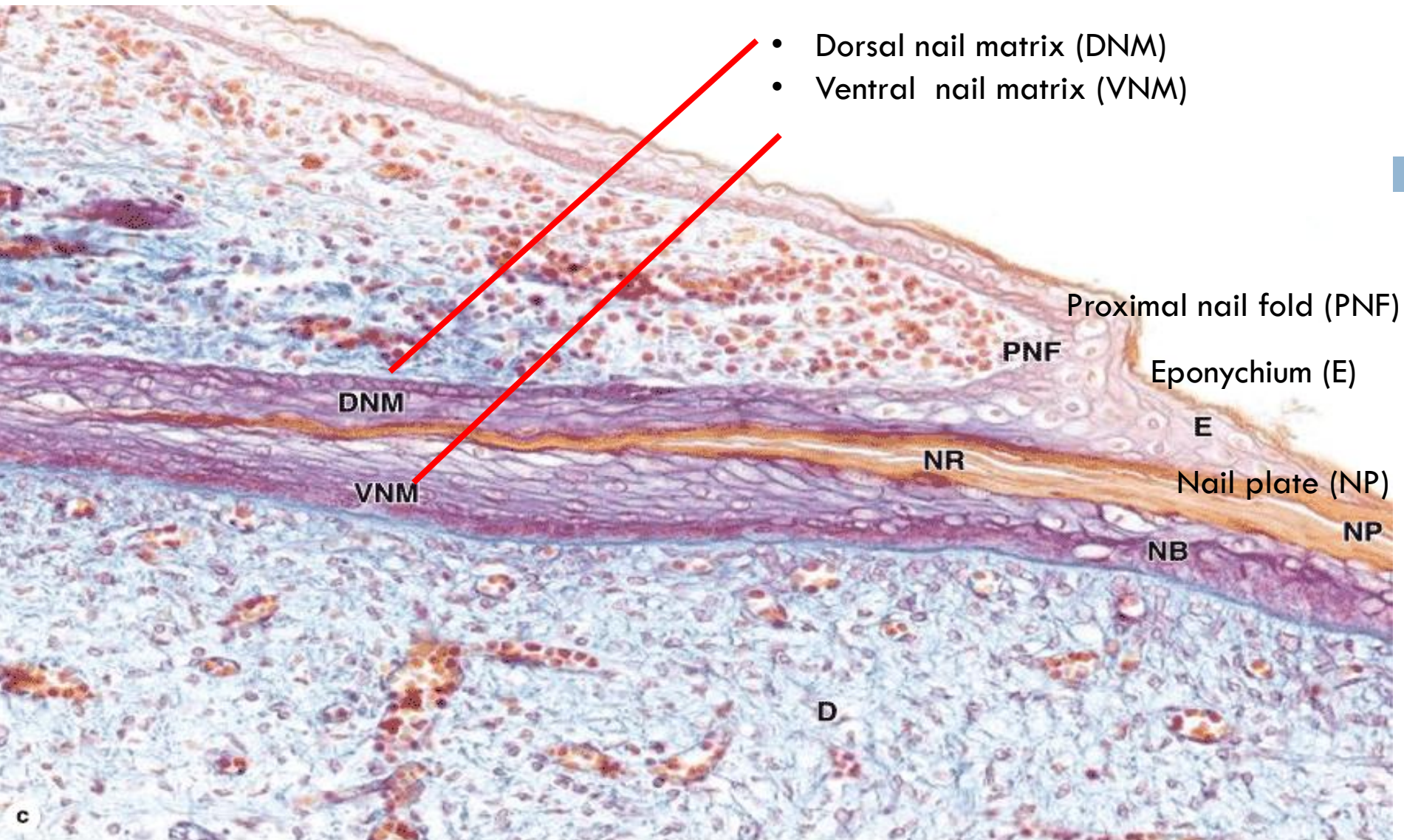
glassy membrane (G)



Blood vessel (BV)
collagen bundles (CB)
Glassy membrane (G)



- Dorsal nail matrix (DNM)
- Ventral nail matrix (VNM)





Glands of the Skin

Sebaceous Glands

- Sebaceous glands are embedded in the **dermis** over most of the body surface, except the thick, hairless (**glabrous**) skin of the palms and soles.
- There is an average of about 100 such glands per square centimeter of skin.
- But the frequency increases to 400–900/cm² in the face and scalp.
- Sebaceous glands are **branched acinar glands**.

Sebaceous Glands

- Several acini converging at a short duct which usually empties into the upper portion of a hair follicle.
- The bulge region of the follicle is a stem cell niche generating cells of the:
 - Hair follicle and matrix
 - Neighboring epidermis
 - Associated sebaceous glands
- In certain hairless regions, such as the genital glans, eyelids, and nipples, sebaceous ducts open directly onto the **epidermal surface**.

Sebaceous Glands

- The acini consist of a basal layer of undifferentiated flattened epithelial cells on the basal lamina.
- These cells proliferate and are displaced toward the middle of the acinus.
- These cells undergo terminal differentiation as distinctly large, lipid-producing **sebocytes** which have their cytoplasm filled with small fat droplets .

Sebaceous Glands

- Their nuclei shrink and undergo autophagy along with other organelles.
- Near the duct the cells disintegrate and release the lipids via holocrine secretion.
- The product of this process is **sebum**, which is gradually moved to the surface of the skin along the **hair follicle or duct**.

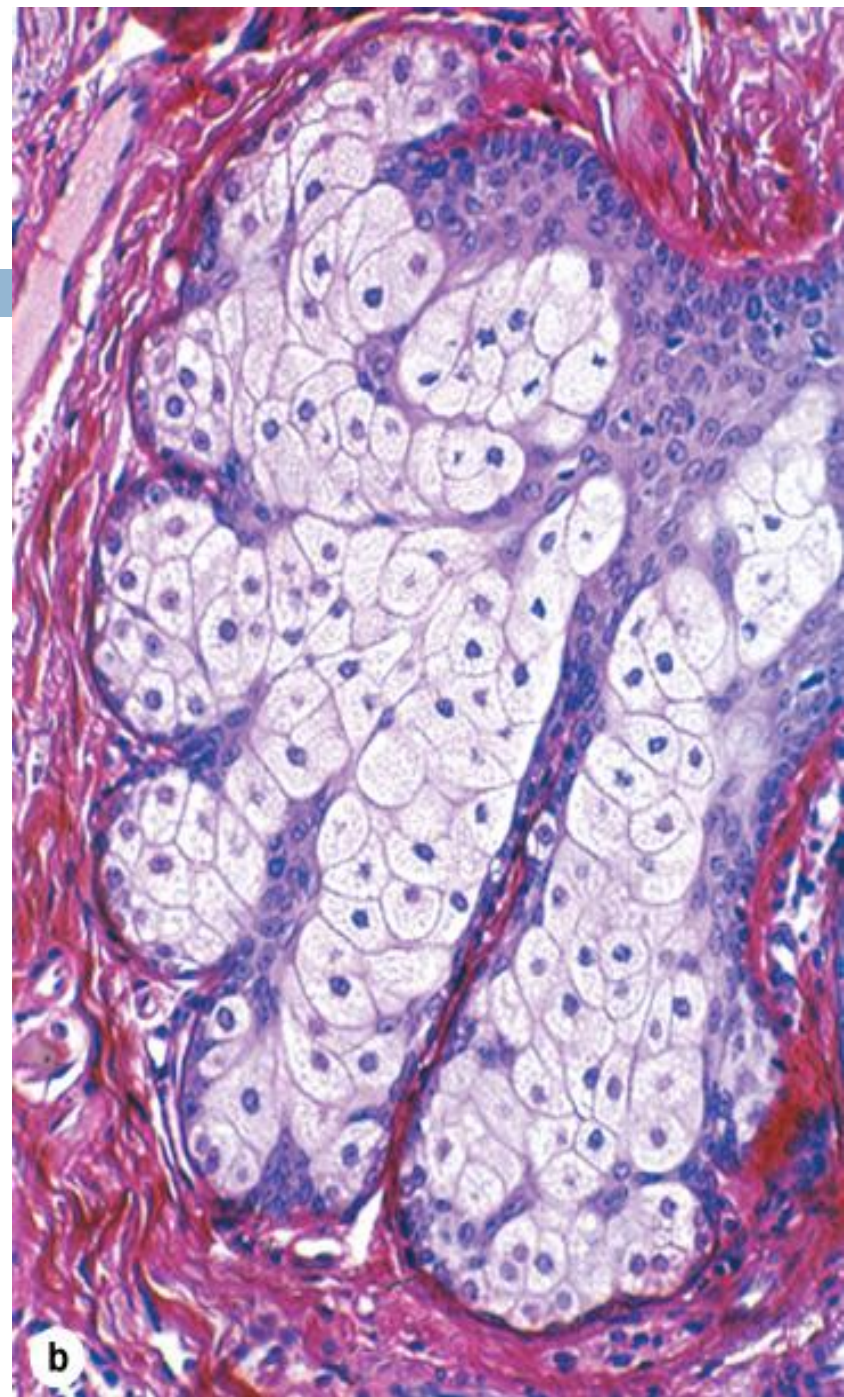
Sebum

- Is a complex mixture of lipids that includes:
 - Wax esters
 - Squalene -- a hydrocarbon that is an intermediate in the formation of cholesterol
 - Cholesterol
 - Triglycerides
 - which are hydrolyzed by bacterial enzymes after secretion

Sebum

- Secretion from sebaceous glands greatly increases at puberty, stimulated primarily by:
 - ▣ testosterone in men and
 - ▣ by ovarian and adrenal androgens in women

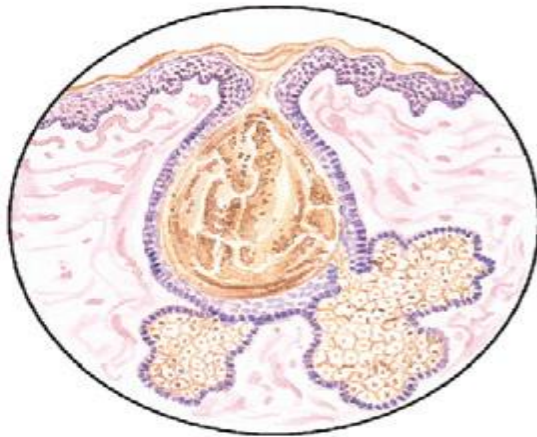
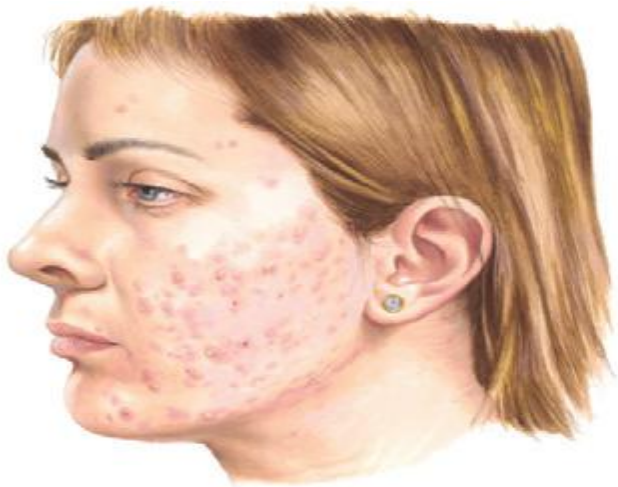
- Specific functions of sebum appear to include:
 - ▣ helping maintain the stratum corneum and hair
 - ▣ exerting weak antibacterial and antifungal properties on the skin surface.



Sebaceous glands.

MEDICAL APPLICATION

- The flow of sebum is continuous, and a disturbance in the normal secretion and flow of sebum is one of the reasons for the development of **acne**.
- **Acne** / áknee/ is a chronic inflammation of obstructed sebaceous glands common during and after puberty.



Acne vulgaris. Clinical manifestation (**Left**) and histologic section (**Right**) showing distended follicle and keratin plug blocking sebum outflow.

Sweat Glands



Sweat glands



- Widely distributed in skin
- Also called sudoriferous glands
 - ▣ Sudor = Sweat
- More than 2.5 million per person



Sweat glands



- Different types
 - ▣ Prevent overheating of the body
 - ▣ Secrete cerumen (ear wax)
 - ▣ Secretion of milk



Sweat glands

- Two main types:
 - Eccrine
 - Apocrine



Sweat Glands

- Sweat glands are epithelial derivatives embedded in the dermis which open to the skin surface or into hair follicles.
- Eccrine sweat glands and apocrine sweat glands have different:
 - distributions
 - functions, and
 - structural details

Eccrine sweat glands

- are widely distributed in the skin and are most numerous on the soles of the feet ($620/\text{cm}^2$).
- Collectively the 3 million eccrine sweat glands of the average person roughly equal the mass of a kidney.
- They can produce as much as 10 L sweat/day, a secretory rate far exceeding that of other **exocrine glands**.
-
- Sweating is the physiological response to increased body temperature during physical exercise or thermal stress.
-
- In humans sweating is the most effective means of temperature regulation.
- Both the secretory portions and ducts of eccrine sweat glands are coiled and have small lumens.

The secretory part of sweat Gland

- Is generally more pale-staining than the ducts and has **stratified cuboidal epithelium consisting of three cell types.**
- Pale pyramidal or columnar **clear cells** :
 - ▣ Produce the sweat
 - ▣ Have abundant mitochondria and
 - ▣ Have microvilli to provide large surface areas
- Interstitial fluid from the capillary-rich dermis around the gland is transported through the clear cells, either:
 - ▣ Directly into the lumen or
 - ▣ Into intercellular canaliculi that open to the lumen.

The secretory part of sweat Gland

□ Pyramidal **dark cells:**

- line most of the luminal surface
- do not touch the basal lamina
- are mucoid and filled with glycoprotein-containing granules
- functions of granules are not well-understood
- But it include components of innate immunity with bactericidal activity.

□ **Myoepithelial cells:**

- **Located** on the basal lamina
- Produce contractions that help discharge secretion into the duct

The ducts of eccrine sweat Glands

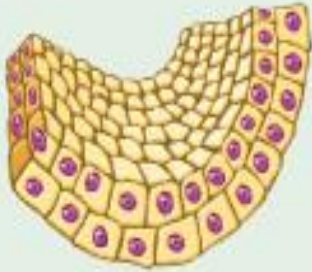
- Consist of two layers of:
 - ▣ more **acidophilic epithelial** cells
 - ▣ The cells are filled with mitochondria
 - ▣ The cells have membranes rich in Na^+ , K^+ -ATPase

The ducts of eccrine sweat Glands

- These duct cells absorb Na^+ ions to prevent excessive loss of this electrolyte.
- After its release on the surface of the skin sweat evaporates, cooling the skin.
- Sweat glands also function as an **auxiliary excretory organ**.
- They eliminate small amounts of nitrogenous waste and excess salts.

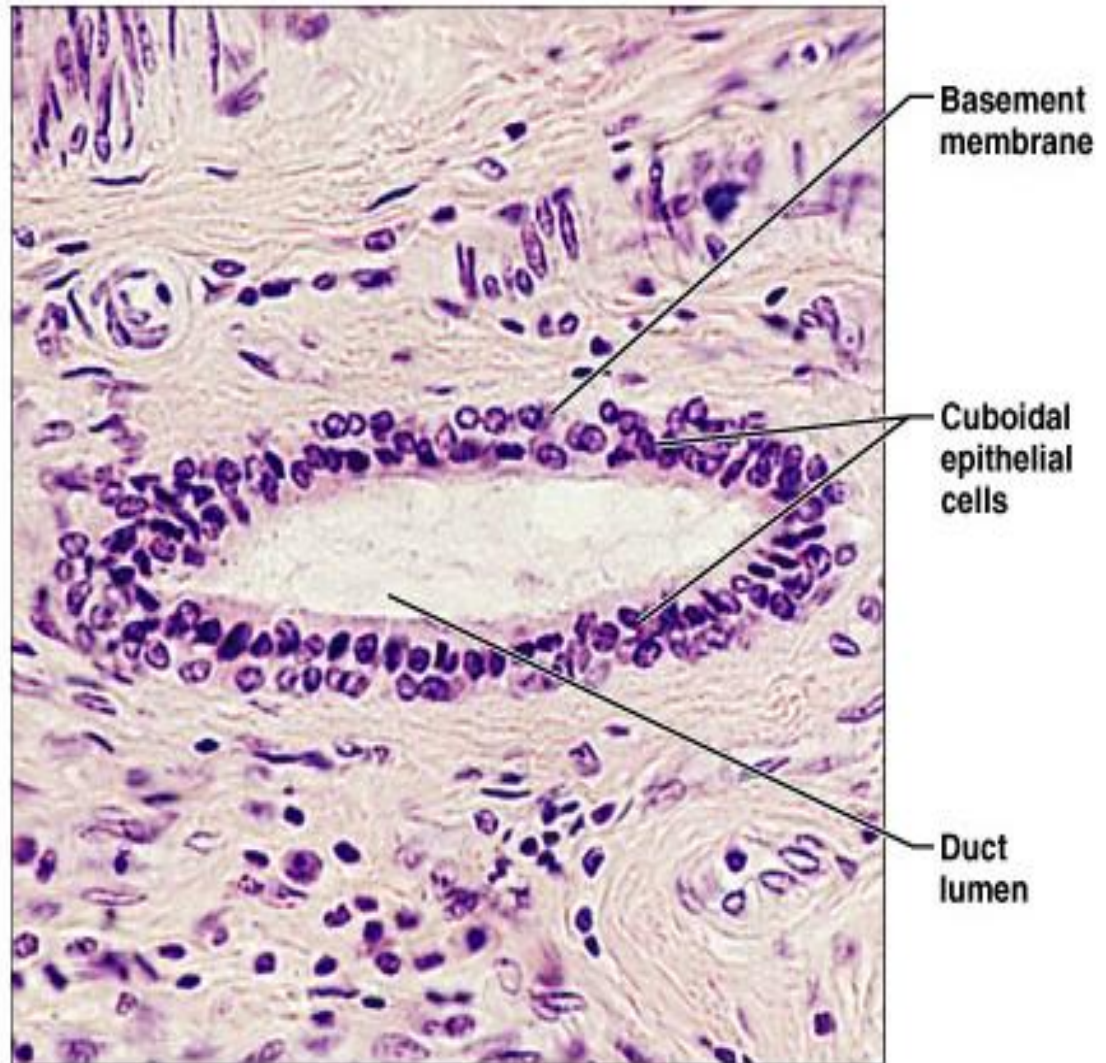
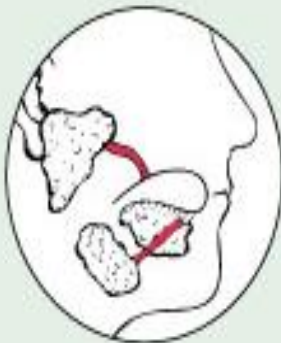
(f) Stratified cuboidal epithelium

Description: Generally two layers of cube-like cells.

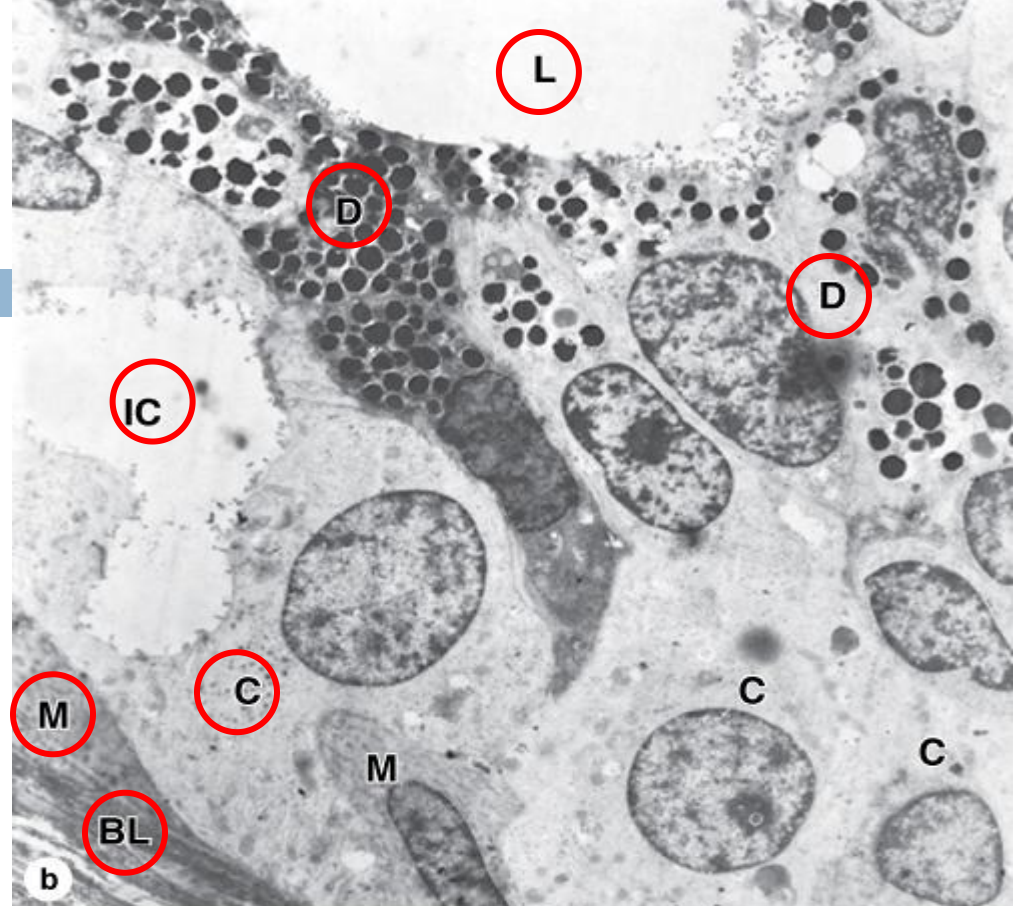
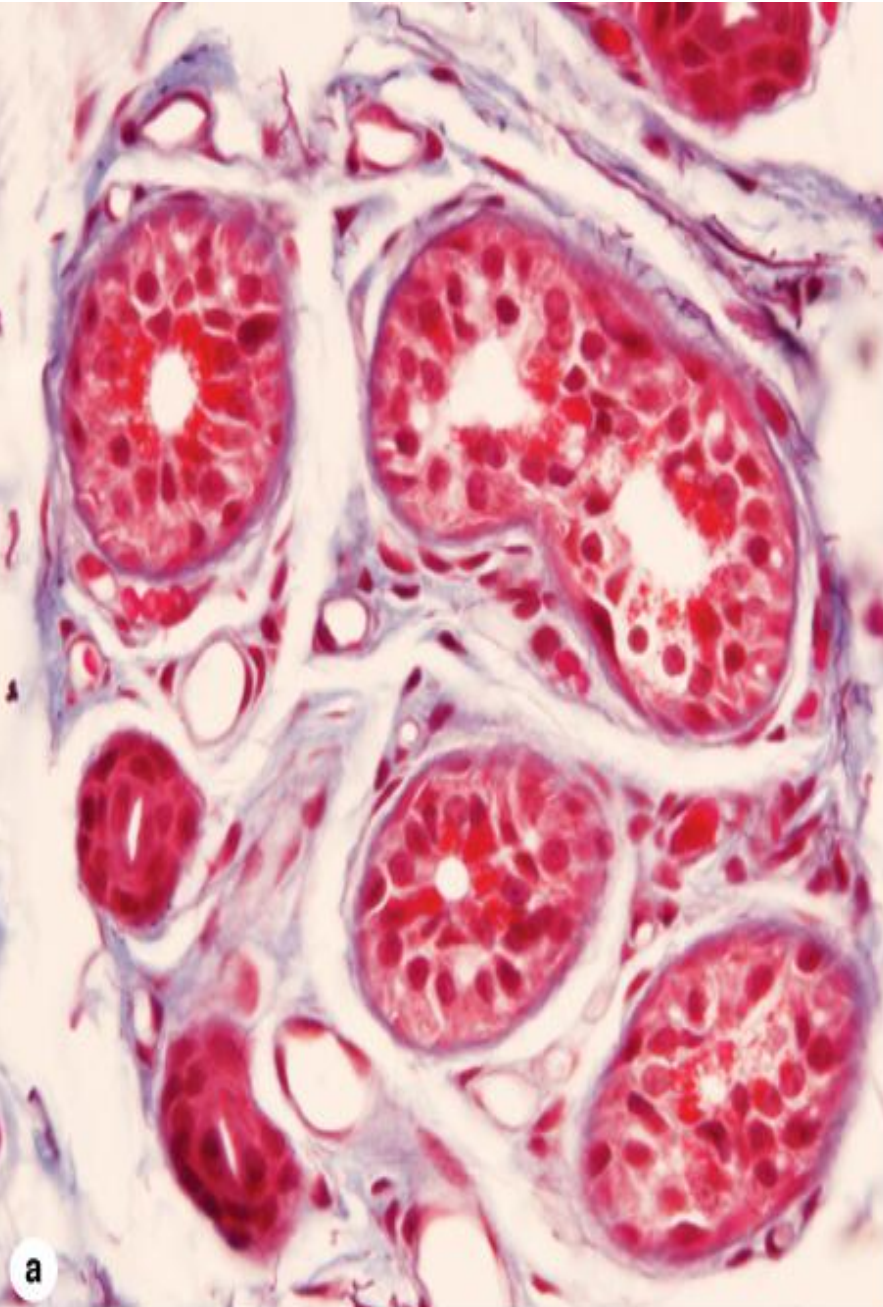


Function: Protection

Location: Largest ducts of sweat glands, mammary glands, and salivary glands.

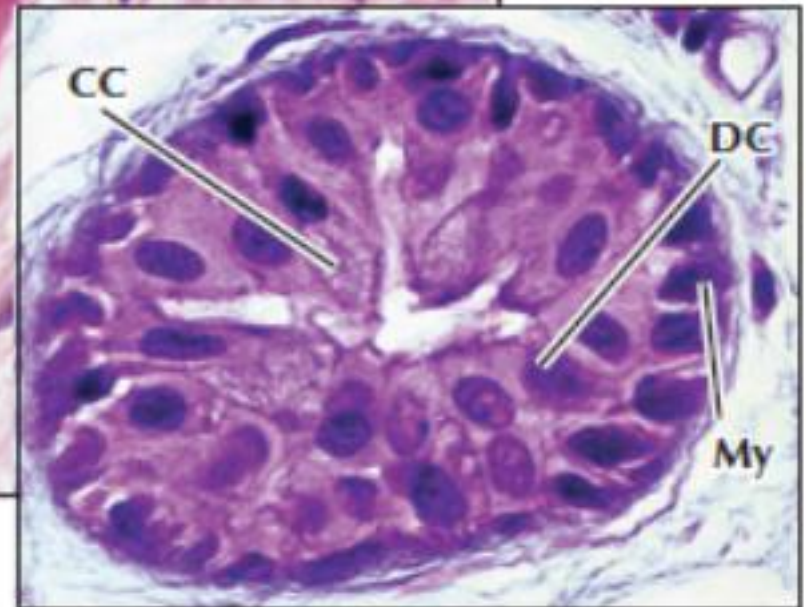
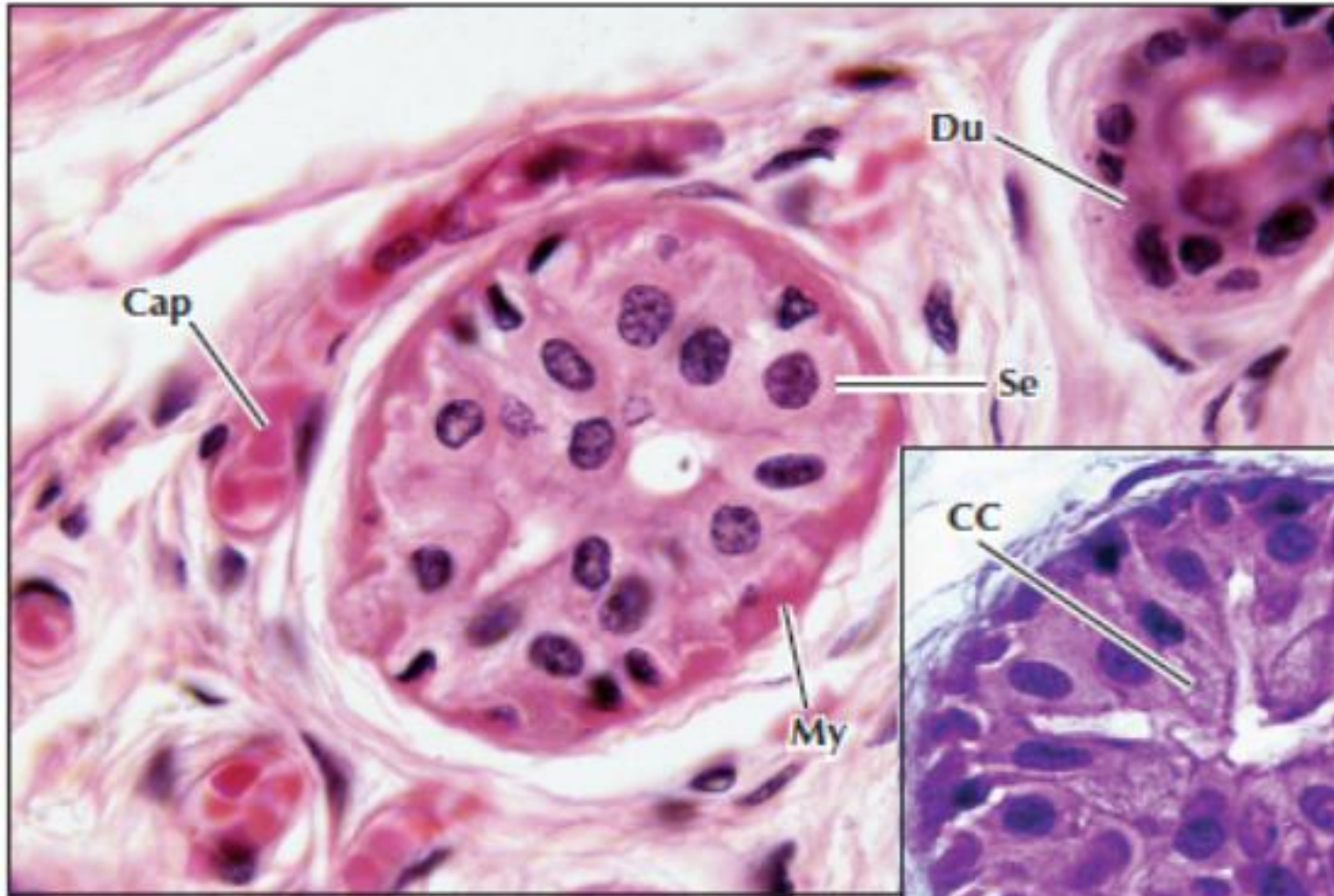


Photomicrograph: Stratified cuboidal epithelium forming a salivary gland duct (300 $\times$).



- Basal lamina (BL)
- Myoepithelial cells (M)
- Clear cells (C)
- Irregular pyramidal cells called dark cells (D)
- Intercellular canaliculi (IC)
- Lumen (L)

a., Eccrine sweat gland secretory cells. b. TEM



secretory cells (**Se**) small duct (**Du**)

myoepithelial cells (**My**)

LM of an acinus of a sweat gland. This staining method distinguishes dark cells (**DC**) from clear cells (**CC**) in the secretory acinus. Surrounding myoepithelial cells (**My**) at the base of the acinus share a basement membrane with secretory cells.

Apocrine sweat glands

- Are largely confined to skin of the axillary and perineal regions.
- Their development (but not functional activity) depends on sex hormones and is not complete until puberty.
- The most obvious histological difference between the two kinds of sweat glands is the much larger lumen of **apocrine glands**.
- The wall of the ducts is similar to that of the eccrine glands.
- The secretory portions of apocrine sweat glands consist of **simple cuboidal, eosinophilic cells**.
- These cells have numerous apical secretory granules that undergo exocytosis.
- Thus the glands are misnamed: their cells show **merocrine, not apocrine, secretion**.

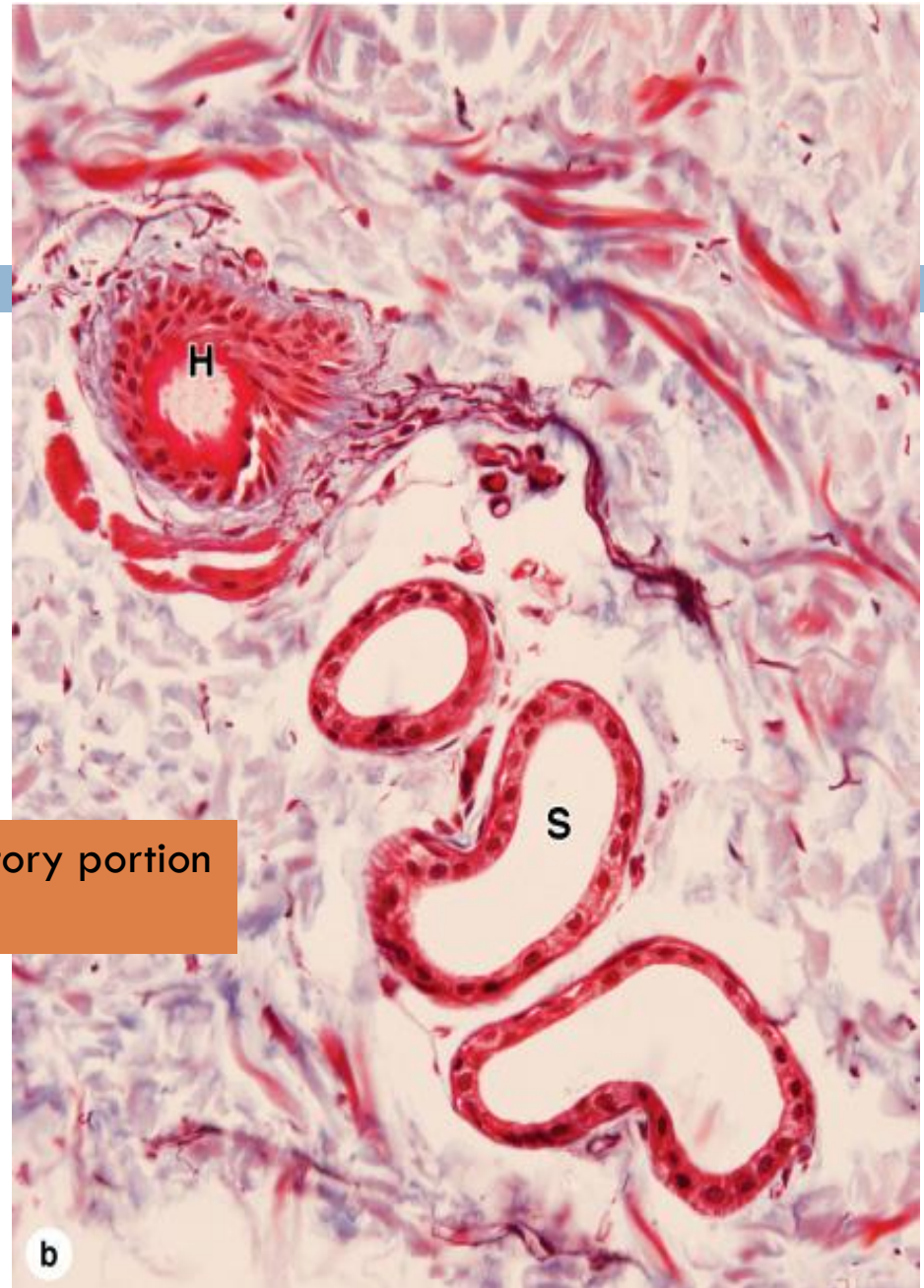
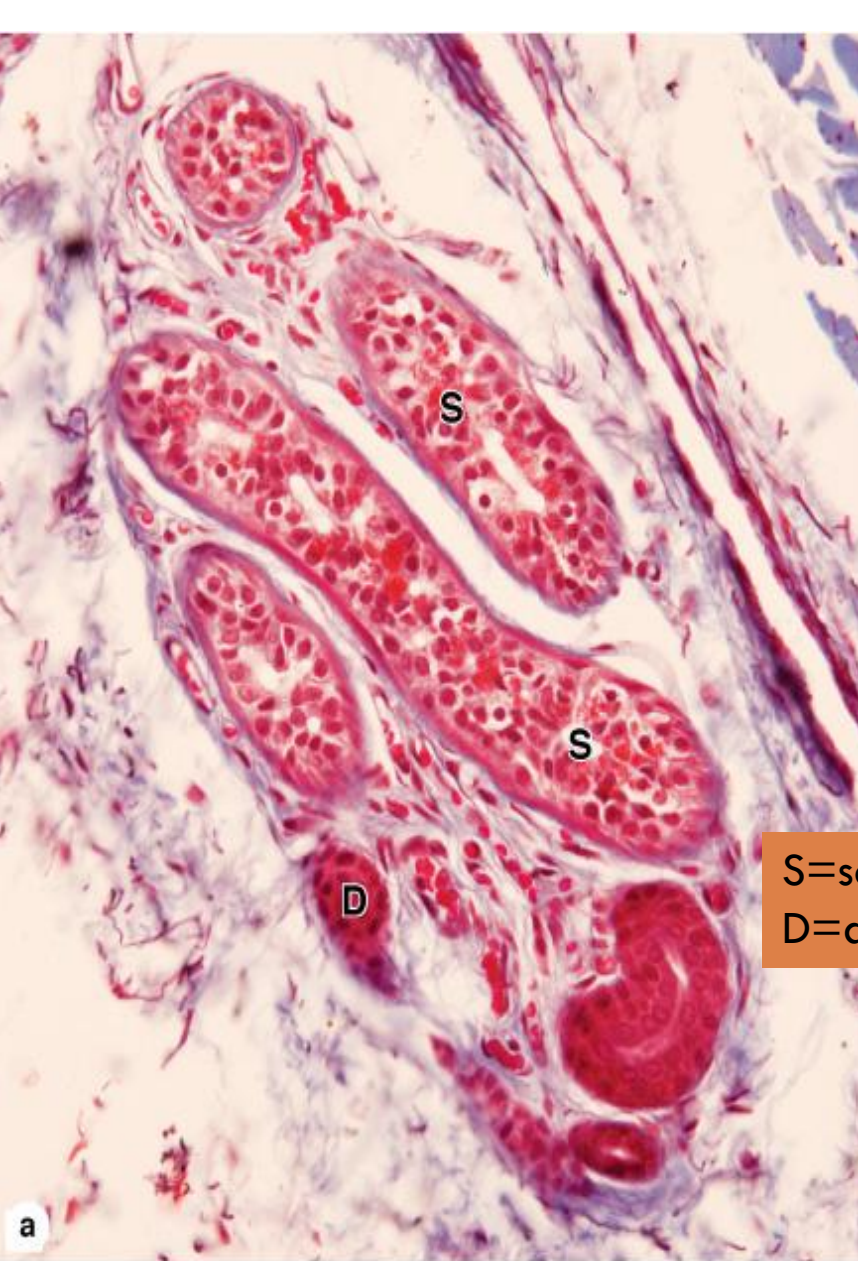
Apocrine sweat glands



- Lumens of apocrine glands often show stored, **protein-rich product**.
- The myoepithelial cells help move the secretion into ducts opening into hair follicles.
- The slightly viscous secretion is initially odorless but may acquire a distinctive odor as a result of bacterial activity.

Apocrine sweat glands

- The production of pheromones by apocrine glands is well-established in many mammals and likely in humans, although in a reduced or vestigial capacity.
- Apocrine sweat glands are innervated by **adrenergic nerve endings/sympa.**
- The eccrine sweat glands receive **cholinergic fibers/para.**



S=secretory portion
D=duct

a. Eccrine and b. apocrine sweat glands.

Tattoos

- Using a needle to deposit pigment in the skin dermis
- Originated around 8000 BC



- Can be dangerous
 - ▣ Risk of allergic reaction and bloodborne infectious diseases such as hepatitis





Thank you